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MIL-M-28787/294B
29 September 1993
SUPERSEDING
MIL-M-28787/294A(EC)
2 August 1983

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

GATE, NAND, DIGITAL, KEY CODE CHP

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the modules described herein shall consist of this specification and the latest issue of MIL-M-28787.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic NAND gate module.

1.2 Classification. This specification covers two classes of modules. Classes I and II as specified in MIL-STD-1389.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Code 603, Bldg. 2917, NAVSURFWARCENDIV, 300 Highway 361, Crane, IN 47522-5001, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. The following specification and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Qualification. Modules furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list by the time of award of contract (see 4.5).

3.2 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.3 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.4 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 0.059 failures per million operating hours for class I or 0.088 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, the failure rate for class I shall be used for reliability prediction (see 6.3).

3.5 Performance requirements. The module shall conform to the electrical requirements specified in table I. Class I modules shall conform to the electrical requirements of class I only, and class II modules shall conform to the electrical requirements of class I and class II (see 6.3).

3.6 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.6.1 Key code. The module key code shall be CHP.

3.6.2 Size. The module size shall be 1A.

3.6.3 Weight. The module shall have a maximum weight of 35 grams.

3.6.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.6.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.6.6 Depth. The module shall have no additional depth requirements.

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3.6.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.6.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part or identifying number M28787/294-1 for class I and M28787/294-2 for class II. No marking shall interfere with heat dissipation from the fin (such as affixing a label which may act as a heat barrier).

3.7 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.8 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 in table I.

3.9 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits in table I (see 6.2).

3.10 Power supply voltage. The module shall be capable of conforming to the requirements in table I for a power supply voltage (VCC) of 5.0 ± 0.5 Vdc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 Vdc ± 1 percent.

4. QUALITY ASSURANCE PROVISIONS

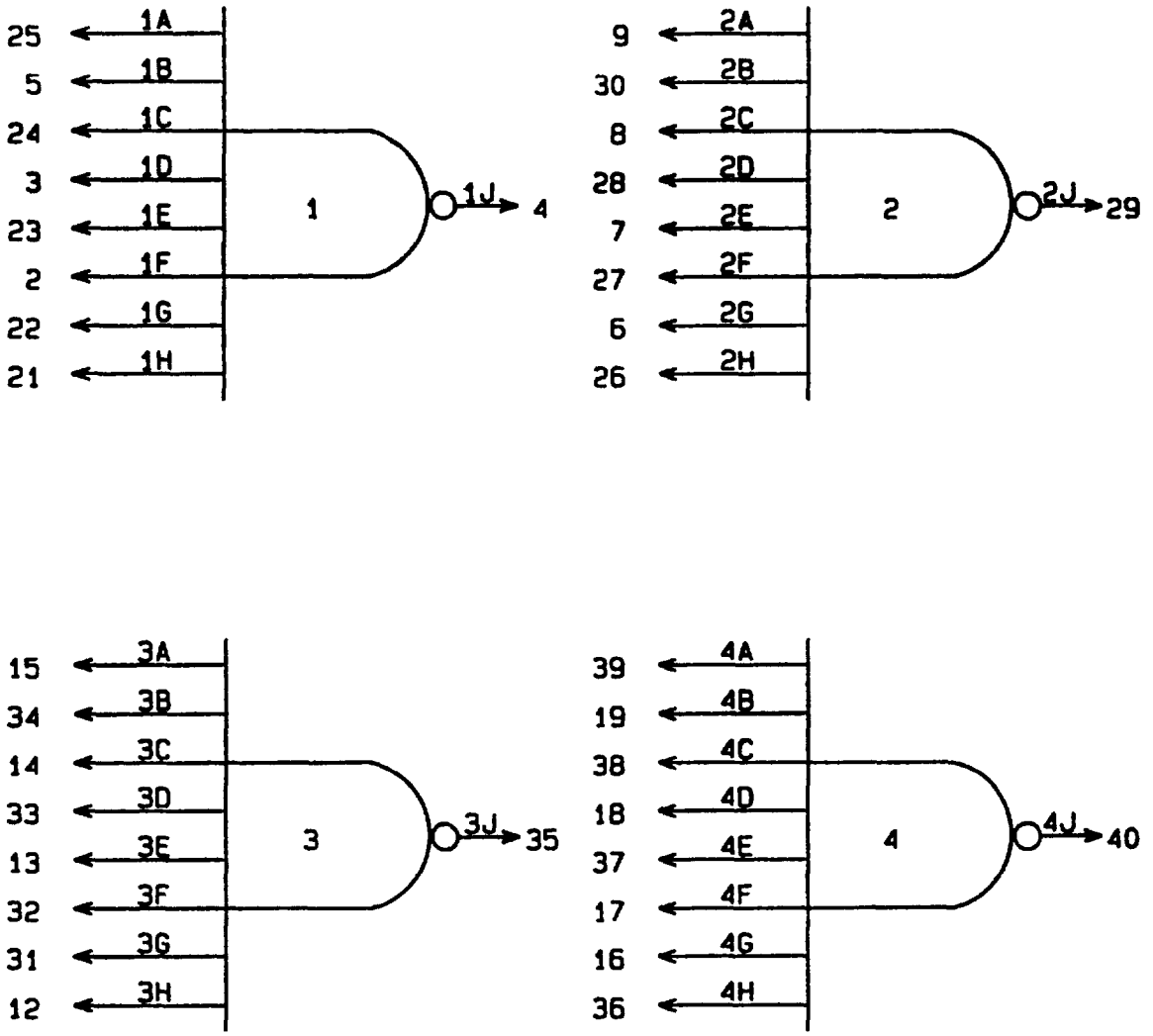
4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. Unless otherwise specified, all resistors shall be metal film and all capacitors shall be ceramic.

4.4 Test procedures. The module shall be tested as specified in table I.

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NOTES:
1. 1A-1H, 2A-2H, 3A-3H, 4A-4H = Data inputs.
2. 1J-4J = Data outputs.
3. Pins 11 and 20 are not used.

FIGURE 1. CHP module function diagram, NAND gate.

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*TABLE I. Module electrical requirements.

Test cond	Parameter	Ref test table	Test type 1/-3/	25°C±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C±5°C		60°C±5°C		-55°C±5°C		85°C±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A1-A3	Isolation Initial End-of-life	 III III	 S S	 10 1	 INF INF	 10 1	 INF INF	 10 1	 INF INF	 10 1	 INF INF	 10 1	 INF INF	 MOhm MOhm
B1-B4	VOUT (low loaded)	IV	S	10	800	10	800	10	800	10	800	10	800	mV
C1-C32	Input breakdown current	V	S	0	100	0	100	0	100	0	100	0	100	µA
D1-D4	Output short circuit current	VI	S	-15	-100	-15	-100	-15	-100	-15	-100	-15	-100	mA
E1-E4	Output leakage current	VII	S	0	1	0	1	0	1	0	1	0	1	mA
F1-F32	Input clamp diode voltage	VIII	S	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	-1.5	V
G1-G4	Output voltage, low level	IX	X	0	400	0	400	0	400	0	400	0	400	mV
H1-H32	Output voltage, high level	X	X	2.5	4 5	2.5	4.5	2.5	4.5	2.5	4.5	2.5	4.5	V
J1-J32	Input current, high level	XI	X	0	20	0	20	0	20	0	20	0	20	µA
K1-K32	Input current, low level	XII	X	-10	-434	-10	-434	-10	-434	-10	-434	-10	-434	µA
L1	Power supply current, low level outputs	XIII	S	0.1	4.4	0.1	4.4	0.1	4.4	0.1	4.4	0.1	4.4	mA

- See footnotes at end of table.

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*TABLE I. Module electrical requirements - Continued.

Test cond	Parameter	Ref test table	Test type 1/-3/	25°C±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C±5°C		60°C±5°C		-55°C±5°C		85°C±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
L2	Power supply current, high level outputs	XIII	S	0.1	2.0	0.1	2.0	0.1	2.0	0.1	2.0	0.1	2.0	mA
M1-M4	Transition time (tTLH), CL = 50 pF	XIV	S	5	50	5	50	5	55	5	50	5	60	ns
M5-M8	Transition time (tTLH), CL = 150 pF	XIV	Q	10	80	10	80	10	90	10	80	10	100	ns
N1-N4	Transition time (tTHL), CL = 50 pF	XIV	S	3	18	3	18	3	20	3	18	3	22	ns
N5-N8	Transition time (tTHL), CL = 150 pF	XIV	Q	6	40	6	40	6	42	6	40	6	44	ns
P1,P9, P17,P25	Propagation delay time (tPLH), CL = 50 pF	XIV	S	3	25	3	25	3	25	3	25	3	27	ns
P2-P8, P10-P16, P18-P24, P26-P32	Propagation delay time (tPLH), CL = 50 pF	XIV	Q	3	25	3	25	3	25	3	25	3	27	ns
P33-P64	Propagation delay time (tPLH), CL = 150 pF	XIV	Q	4	35	4	35	4	35	4	35	4	37	ns
Q1,Q9 Q17,Q25	Propagation delay time (tPHL), CL = 50 pF	XIV	S	3	35	3	35	3	35	3	35	3	37	ns

- See footnotes at end of table.

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*TABLE I. Module electrical requirements - Continued.

Test cond	Parameter	Ref test table	Test type 1/-3/	25°C±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C±5°C		60°C±5°C		-55°C±5°C		85°C±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
Q2-Q8, Q10-Q16, Q18-Q24, Q26-Q32	Propagation delay time (tPHL), CL = 50 pF	XIV	Q	3	35	3	35	3	35	3	35	3	37	ns
Q33-Q64	Propagation delay time (tPHL), CL = 150 pF	XIV	Q	4	45	4	45	4	45	4	45	4	47	ns
R1-R4	Output crosstalk, low level	XV	Q	0	500	0	500	0	500	0	500	0	500	mV
S1-S4	Output crosstalk, high level	XV	Q	0	400	0	400	0	400	0	400	0	400	mV
T	Power supply transient amplitude	XVI	Q	0	400	0	400	0	400	0	400	0	400	mV
U	Power supply filter capacitance	XVII	S	1.6	3.0	1.6	3.0	1.6	3.0	1.6	3.0	1.6	3.0	μF
V1-V32	Input capacitance	XVIII	Q	5	15	5	15	5	15	5	15	5	15	pF

1/ "X" parameters shall be 100 percent tested during the module acceptance test.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.
4/ Unless otherwise specified, limits are for initial and end-of-life.

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4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Quality Assurance Activity (SEM-QAA) and they conform to the requirements of MIL-M-28787.

* 4.4.2 Order of testing. Test conditions A through V shall be performed in any order with the following exceptions. Test conditions A, B, C, D, E, and F are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, B, C, D, E or F are part of a sequence of tests, the order of testing shall be as follows: A, B, C, D, E, and F where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

* 4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through V shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ± 5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions G through U shall be performed in the order specified in 4.4.2 using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests in table I which are marked with an "X" in the test type column, and test condition U, shall be performed using the initial limits at 25°C ± 5°C ambient.

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* 4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests G through U shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table I. Following the return of the module to 25°C, electrical tests G through V shall be performed in the order specified in 4.4.2 using the 25°C ± 5°C ambient end-of-life requirements in table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements in table I, the module is considered to have completed its useful life (see 3.9).

6.3 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in a class I environment, the failure rate for class I shall be used for reliability prediction (see 3.3 and 3.4).

6.4 Module description. The following electrical parameters reflect the 0°C to 60°C and -55°C to 85°C end-of-life limits in table I.

	Class I	Class II	Units
Maximum power dissipation (VCC = 5.5 V)			
Outputs low (not loaded)	24.2	24.2	mW
Outputs high (not loaded)	11.0	11.0	mW
Supply voltage	5 ± 10%	5 ± 10%	Vdc
Input voltage			
Low level	0.7 maximum	0.7 maximum	V
High level	2 minimum	2 minimum	V

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Input current			
Low level (VIL = 0 V)	-434 maximum	-434 maximum	µA
Low level (VIL = 0.4 V)	-400 maximum	-400 maximum	µA
High level (VIH = 2.5 V)	20 maximum	20 maximum	µA
High level (VIH = 7 V)	100 maximum	100 maximum	µA
Output voltage			
Low level (IOL = 4 mA)	400 maximum	400 maximum	mV
High level (IOH = -400 uA)	2.5 minimum	2.5 minimum	V
Output drive current			
Low level (VOL = 0.4 V)	4 maximum	4 maximum	mA
High level (VOH = 2.5 V)	-400 maximum	-400 maximum	µA

6.5 Module function. This module contains four independent 8-input NAND gates.

6.5.1 Input to output relationship. Each NAND gate has the input to output relationship in table II for positive logic.

TABLE II. Input to output relationship. 1/ 2/

Inputs								Output
A	B	C	D	E	F	G	H	J
H	H	H	H	H	H	H	H	L
L	H	H	H	H	H	H	H	H
H	L	H	H	H	H	H	H	H
H	H	L	H	H	H	H	H	H
H	H	H	L	H	H	H	H	H
H	H	H	H	L	H	H	H	H
H	H	H	H	H	L	H	H	H
H	H	H	H	H	H	L	H	H
H	H	H	H	H	H	H	L	H
L	L	L	L	L	L	L	L	H

1/ H = High level.
2/ L = Low level.

6.5.2 Parallel outputs. Paralleling of outputs is not permissible.

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6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

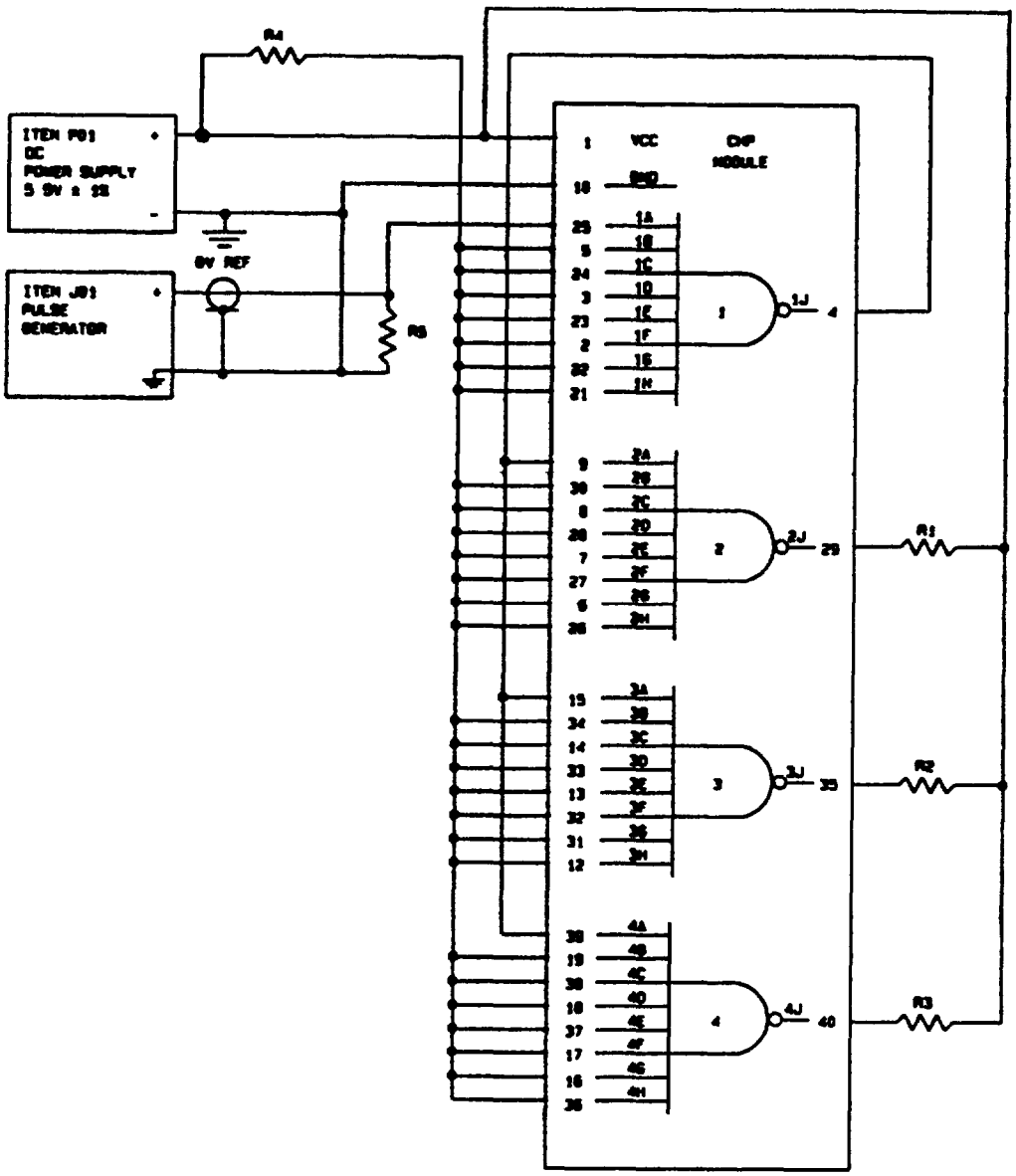
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Navy - NW
Air Force - 85

Review activities:
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Navy - AS, MC
Air Force - 13, 19
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Preparing activity
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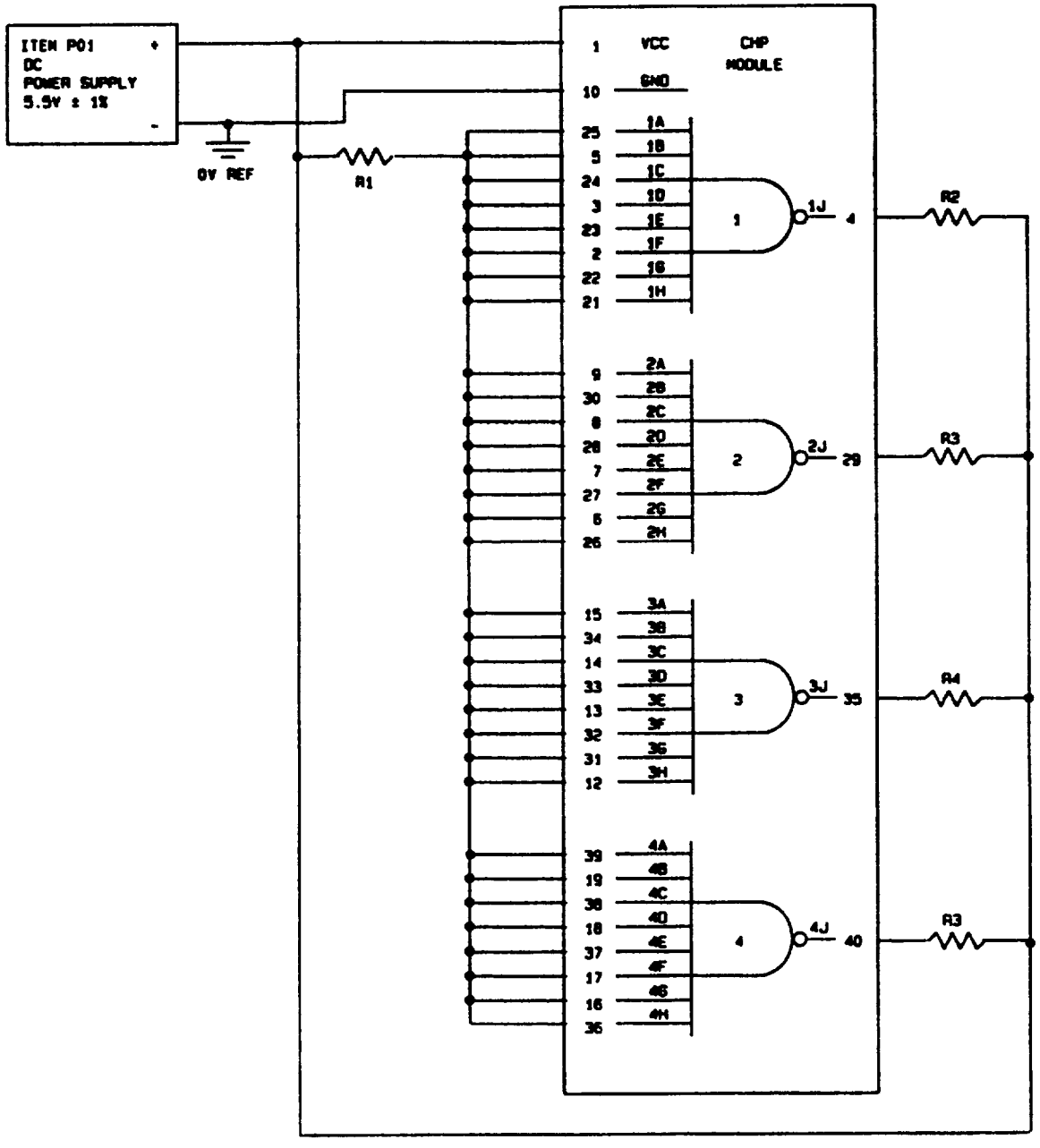
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NOTES:
1. R1-R3 = 1.27 kOhm.
2. R4 = 1 kOhm ± 1%, 1/4 W.
3. R5 = 51 kOhm ± 5%, 1/2 W.
4. Item J01 setup:
   PW = 500 ± 100 ns.
   Frequency = 1 MHz ± 20%.
   tTLH = tTHL = 50 ± 10 ns.
   PA = 3.5 ± 0.5 V.

FIGURE 2. Operating temperature and life test circuit.

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NOTES:
1. R1 = 1 kOhm ± 5%, 1/4 W.
2. R2-R5 = 1.27 kOhm ± 1%, 1/4 W.

FIGURE 3. Component temperature test circuit.

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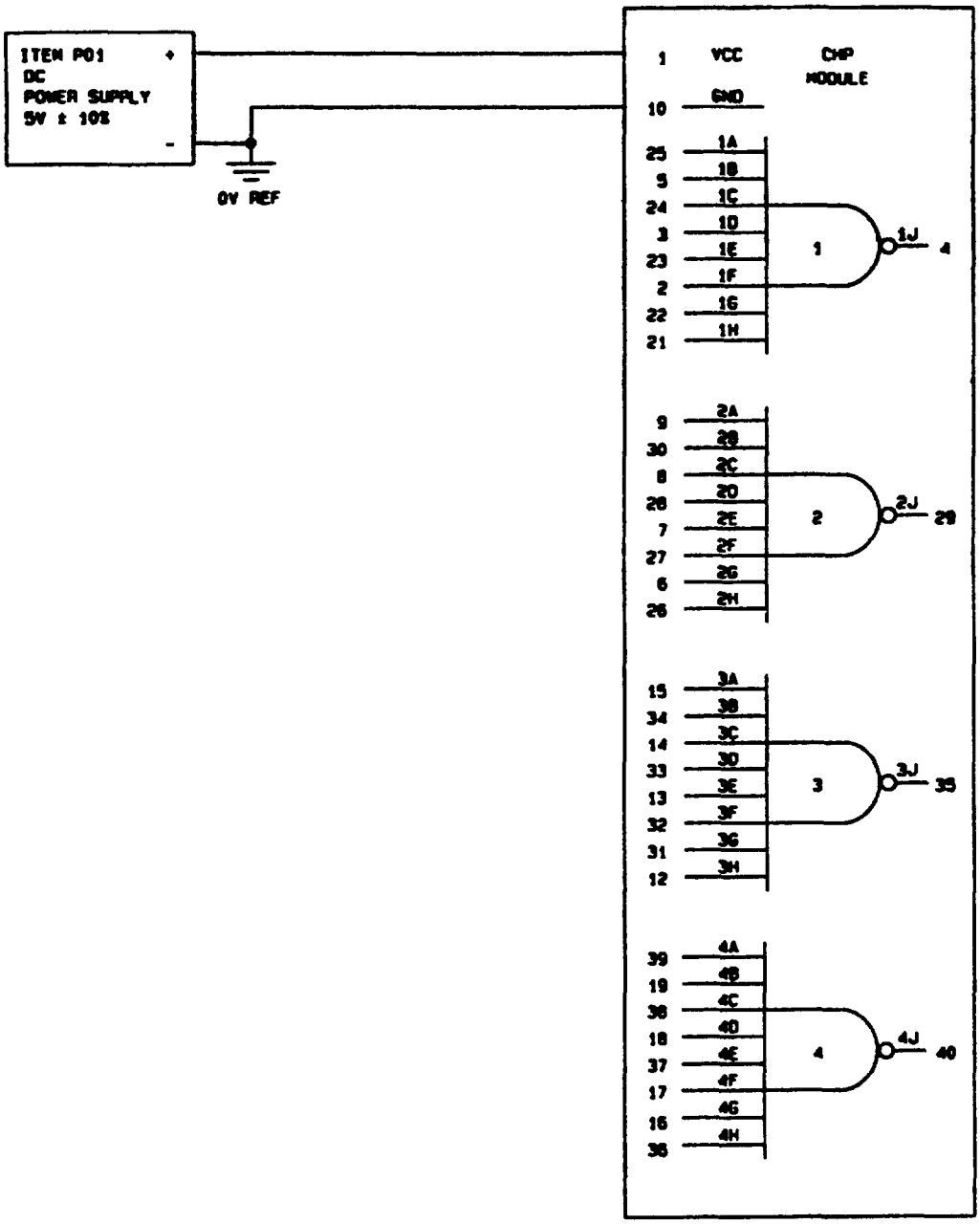


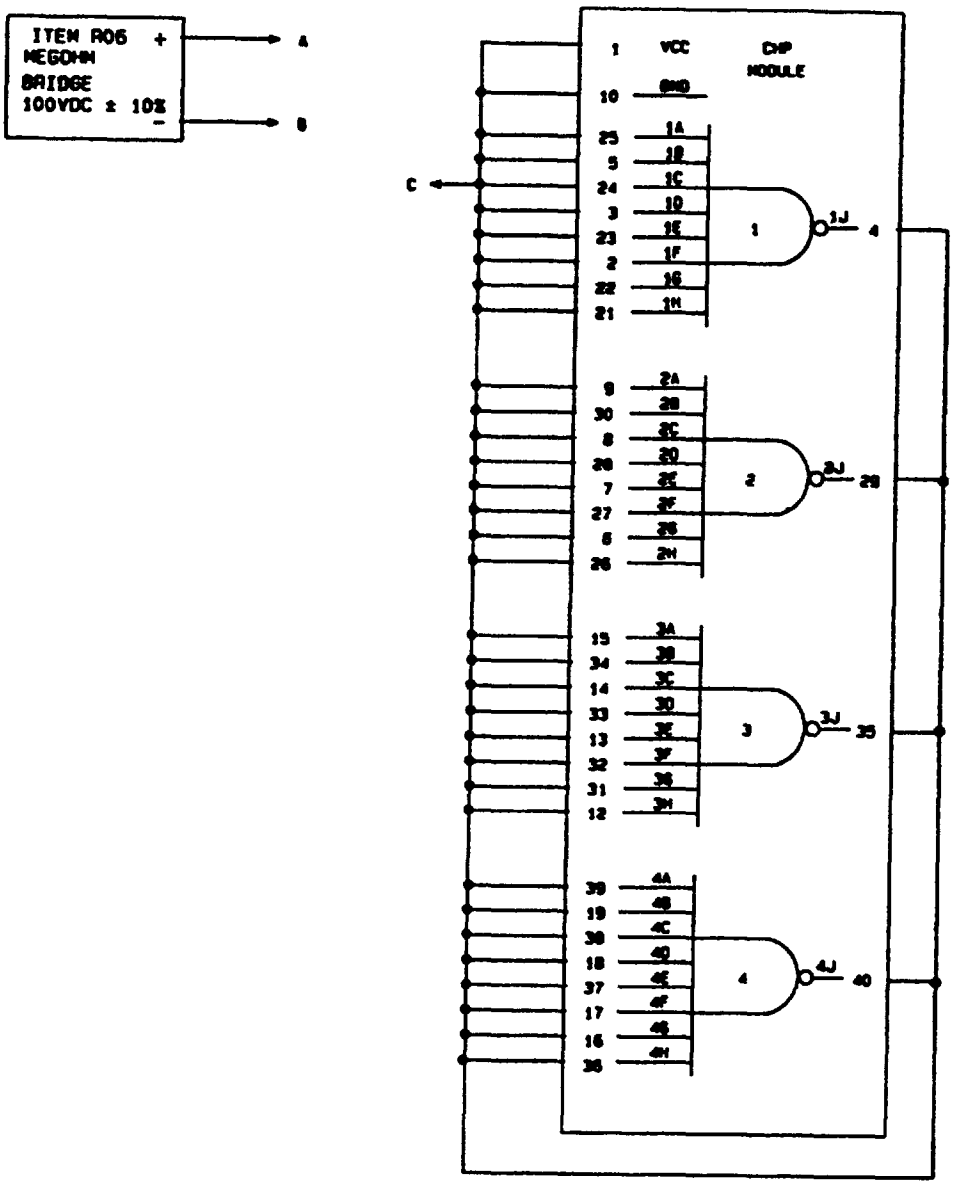
FIGURE 4. Vibration test circuit.

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TABLE III. Isolation test.

Test cond	Parameter	Connect		Initial limits 25°C±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A1	Isolation	C plus 11 and 20	Frame	10	INF	MOhm	5
A2		C	11 and 20				
A3	Isolation	11	20	10	INF	MOhm	5

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WARNING:
When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: For test condition A1, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

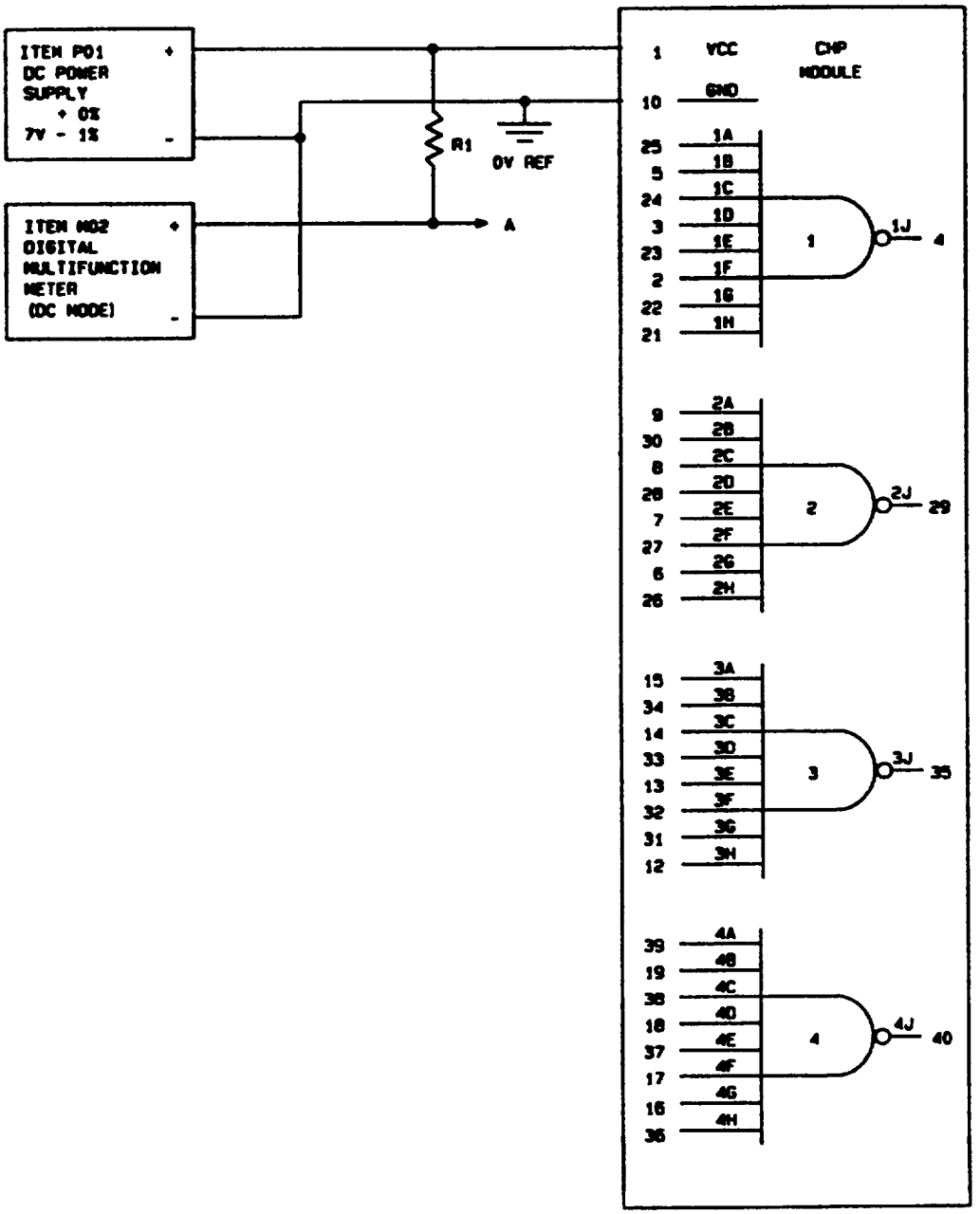
FIGURE 5. Isolation test circuit for table III.

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TABLE IV. VOUT (low-loaded) test.

Test cond	Parameter	Connect A to pin	Initial limits 25°C±5°C ambient Min	Max	Unit	Test fig
B1	1J VOUT (low loaded)	4	10	800	mV	6
B2	2J	29				
B3	3J	35				
B4	4J VOUT (low loaded)	40	10	800	mV	6

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NOTE: R1 = 1.27 kOhm ± 1%, 1/4 W.

FIGURE 6. VOUT (low-loaded) test circuit for table IV.

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TABLE V. Input breakdown current test.

Test cond	Parameter	Connect		Initial limits 25±5°C ambient		Unit	Test fig
		A to pin	B to pin	Min	Max		
C1	1A Input breakdown current	2,3,5,6,7,8, 9,12,13,14, 15,16,17,18, 19,21,22,23, 24,26,27,28, 30,31,32,33, 34,36,37,38, 39	25	0	100	µA	7
C2	1B	2,3,6,7,8,9, 12,13,14,15, 16,17,18,19, 21,22,23,24, 25,26,27,28, 30,31,32,33, 34,36,37,38, 39	5				
C3	1C	2,3,5,6,7,8, 9,12,13,14, 15,16,17,18, 19,21,22,23, 25,26,27,28, 30,31,32,33, 34,36,37,38, 39	24				
C4	1D Input breakdown current	2,5,6,7,8,9, 12,13,14,15, 16,17,18,19, 21,22,23,24, 25,26,27,28, 30,31,32,33, 34,36,37,38, 39	3	0	100	µA	7

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TABLE V. Input breakdown current test - Continued.

Test cond	Parameter	Connect A to pin	Connect B to pin	Initial limits 25±5°C ambient Min	Initial limits 25±5°C ambient Max	Unit	Test fig
C5	1E Input breakdown current	2,3,5,6,7,8, 9,12,13,14, 15,16,17,18, 19,21,22,24, 25,26,27,28, 30,31,32,33, 34,36,37,38, 39	23	0	100	µA	7
C6	1F	3,5,6,7,8,9, 12,13,14,15, 16,17,18,19, 21,22,23,24, 25,26,27,28, 30,31,32,33, 34,36,37,38, 39	2				
C7	1G	2,3,5,6,7,8, 9,12,13,14, 15,16,17,18, 19,21,23,24, 25,26,27,28, 30,31,32,33, 34,36,37,38, 39	22				
C8	1H Input breakdown current	2,3,5,6,7,8, 9,12,13,14, 15,16,17,18, 19,22,23,24, 25,26,27,28, 30,31,32,33, 34,36,37,38, 39	21	0	100	µA	7

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TABLE V. Input breakdown current test - Continued.

Test cond	Parameter	Connect A to pin	Connect B to pin	Initial limits 25±5°C ambient Min	Initial limits 25±5°C ambient Max	Unit	Test fig
C9	2A Input breakdown current	2,3,5,6,7,8, 12,13,14,15, 16,17,18,19, 21,22,23,24, 25,26,27,28, 30,31,32,33, 34,36,37,38, 39	9	0	100	µA	7
C10	2B	2,3,5,6,7,8, 9,12,13,14, 15,16,17,18, 19,21,22,23, 24,25,26,27, 28,31,32,33, 34,36,37,38, 39	30				
C11	2C	2,3,5,6,7,9, 12,13,14,15, 16,17,18,19, 21,22,23,24, 25,26,27,28, 30,31,32,33, 34,36,37,38, 39	8				
C12	2D Input breakdown current	2,3,5,6,7,8, 9,12,13,14, 15,16,17,18, 19,21,22,23, 24,25,26,27, 30,31,32,33, 34,36,37,38, 39	28	0	100	µA	7

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TABLE V. Input breakdown current test - Continued.

Test cond	Parameter	Connect		Initial limits 25±5°C ambient		Unit	Test fig
		A to pin	B to pin	Min	Max		
C13	2E Input breakdown current	2,3,5,6,8,9, 12,13,14,15, 16,17,18,19, 21,22,23,24, 25,26,27,28, 30,31,32,33, 34,36,37,38, 39	7	0	100	µA	7
C14	2F	2,3,5,6,7,8, 9,12,13,14, 15,16,17,18, 19,21,22,23, 24,25,26,28, 30,31,32,33, 34,36,37,38, 39	27				
C15	2G	2,3,5,7,8,9, 12,13,14,15, 16,17,18,19, 21,22,23,24, 25,26,27,28, 30,31,32,33, 34,36,37,38, 39	6				
C16	2H Input breakdown current	2,3,5,6,7,8, 9,12,13,14, 15,16,17,18, 19,21,22,23, 24,25,27,28, 30,31,32,33, 34,36,37,38, 39	26	0	100	µA	7

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TABLE V. Input breakdown current test - Continued.

Test cond	Parameter	Connect		Initial limits 25±5°C ambient		Unit	Test fig
		A to pin	B to pin	Min	Max		
C17	3A Input breakdown current	2,3,5,6,7,8, 9,12,13,14, 16,17,18,19, 21,22,23,24, 25,26,27,28, 30,31,32,33, 34,36,37,38, 39	15	0	100	µA	7
C18	3B	2,3,5,6,7,8, 9,12,13,14, 15,16,17,18, 19,21,22,23, 24,25,26,27, 28,30,31,32, 33,36,37,38, 39	34				
C19	3C	2,3,5,6,7,8, 9,12,13,15, 16,17,18,19, 21,22,23,24, 25,26,27,28, 30,31,32,33, 34,36,37,38, 39	14				
C20	3D Input breakdown current	2,3,5,6,7,8, 9,12,13,14, 15,16,17,18, 19,21,22,23, 24,25,26,27, 28,30,31,32, 34,36,37,38, 39	33	0	100	µA	7

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TABLE V. Input breakdown current test - Continued.

Test cond	Parameter	Connect A to pin	Connect B to pin	Initial limits 25±5°C ambient Min	Initial limits 25±5°C ambient Max	Unit	Test fig
C21	3E Input breakdown current	2,3,5,6,7,8, 9,12,14,15, 16,17,18,19, 21,22,23,24, 25,26,27,28, 30,31,32,33, 34,36,37,38, 39	13	0	100	μA	7
C22	3F	2,3,5,6,7,8, 9,12,13,14, 15,16,17,18, 19,21,22,23, 24,25,26,27, 28,30,31,33, 34,36,37,38, 39	32				
C23	3G	2,3,5,6,7,8, 9,12,13,14, 15,16,17,18, 19,21,22,23, 24,25,26,27, 28,30,32,33, 34,36,37,38, 39	31				
C24	3H Input breakdown current	2,3,5,6,7,8, 9,13,14,15, 16,17,18,19, 21,22,23,24, 25,26,27,28, 30,31,32,33, 34,36,37,38, 39	12	0	100	μA	7

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:26Z
Check the source to verify that this is the current version before use.

TABLE V. Input breakdown current test - Continued.

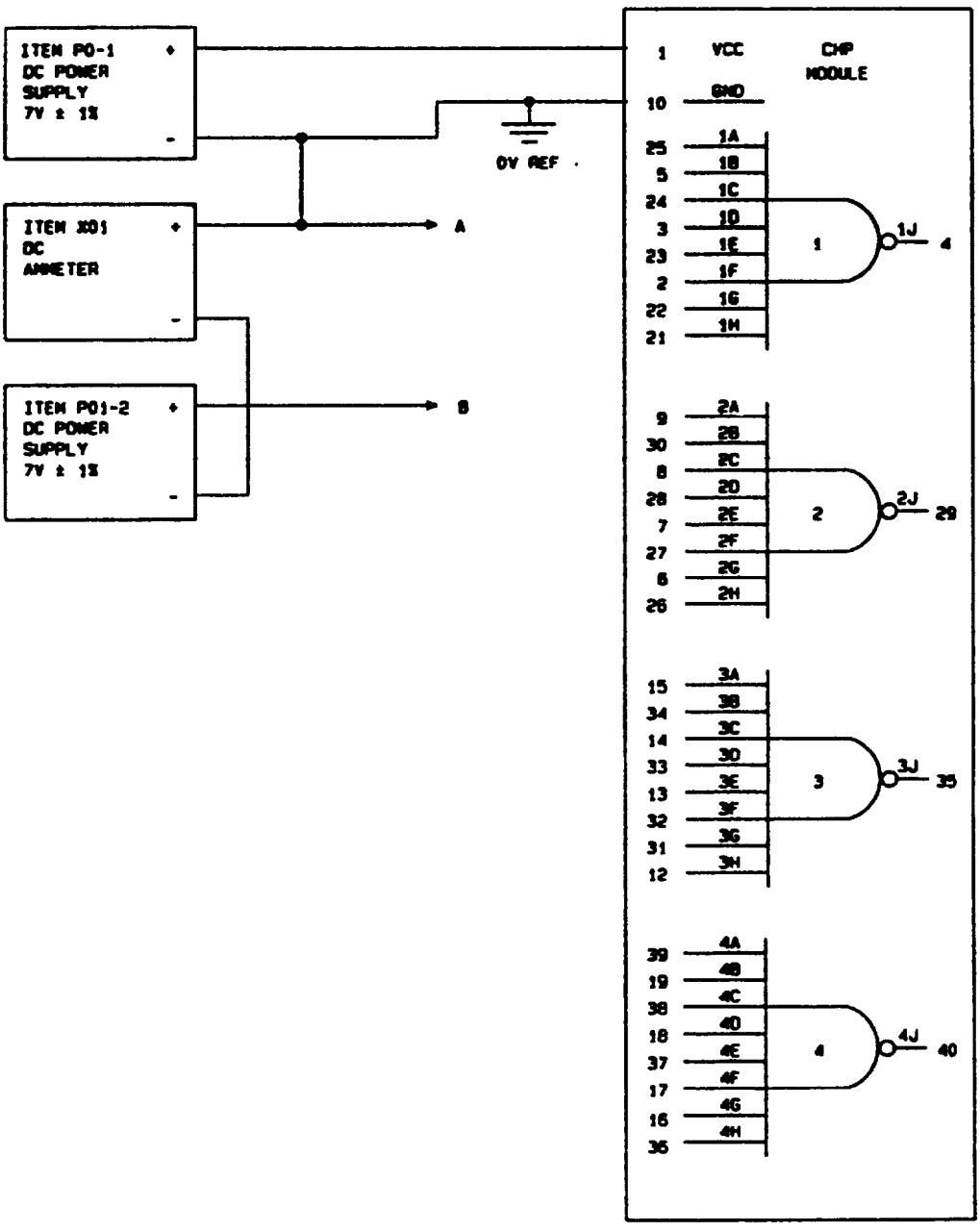
Test cond	Parameter	Connect A to pin	Connect B to pin	Initial limits 25±5°C ambient Min	Initial limits 25±5°C ambient Max	Unit	Test fig
C25	4A Input breakdown current	2,3,5,6,7,8,9,12,13,14,15,16,17,18,19,21,22,23,24,25,26,27,28,30,31,32,33,34,36,37,38	39	0	100	µA	7
C26	4B	2,3,5,6,7,8,9,12,13,14,15,16,17,18,21,22,23,24,25,26,27,28,30,31,32,33,34,36,37,38,39	19				
C27	4C	2,3,5,6,7,8,9,12,13,14,15,16,17,18,19,21,22,23,24,25,26,27,28,30,31,32,33,34,36,37,39	38				
C28	4D Input breakdown current	2,3,5,6,7,8,9,12,13,14,15,16,17,19,21,22,23,24,25,26,27,28,30,31,32,33,34,36,37,38,39	18	0	100	µA	7

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TABLE V. Input breakdown current test - Continued.

Test cond	Parameter	Connect		Initial limits 25±5°C ambient		Unit	Test fig
		A to pin	B to pin	Min	Max		
C29	4E Input breakdown current	2,3,5,6,7,8, 9,12,13,14, 15,16,17,18, 19,21,22,23, 24,25,26,27, 28,30,31,32, 33,34,36,38, 39	37	0	100	µA	7
C30	4F	2,3,5,6,7,8, 9,12,13,14, 15,16,18,19, 21,22,23,24, 25,26,27,28, 30,31,32,33, 34,36,37,38, 39	17				
C31	4G	2,3,5,6,7,8, 9,12,13,14, 15,17,18,19, 21,22,23,24, 25,26,27,28, 30,31,32,33, 34,36,37,38, 39	16				
C32	4H Input breakdown current	2,3,5,6,7,8, 9,12,13,14, 15,16,17,18, 19,21,22,23, 24,25,26,27, 28,30,31,32, 33,34,37,38, 39	36	0	100	µA	7

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CAUTION:
Do not isolate the chassis of item X01 from earth ground.

NOTE: Item P01-2 shall be limited to 5 mA current.

FIGURE 7. Input breakdown current test circuit for table V.

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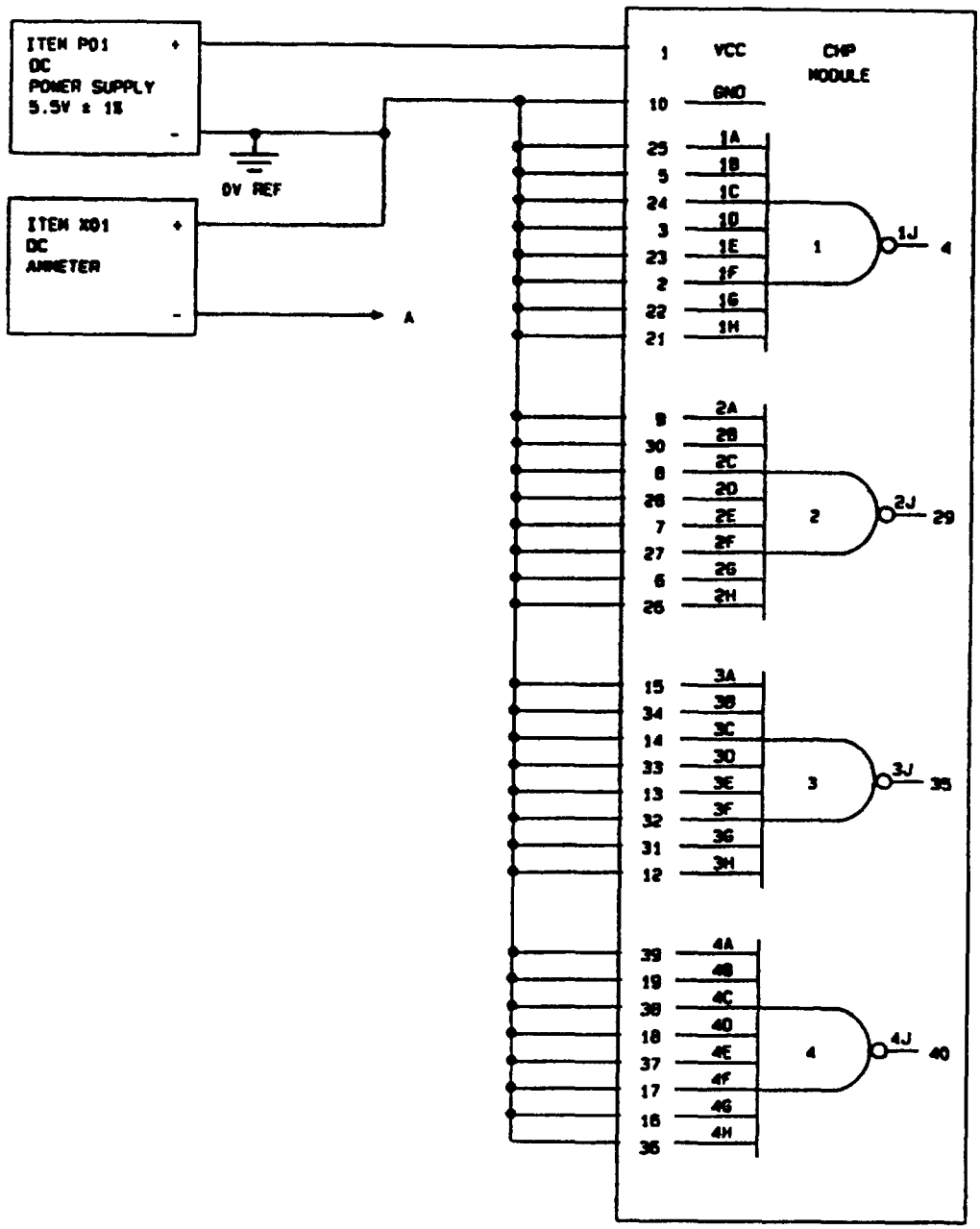
TABLE VI. Output short circuit current test.

Test cond	Parameter	Connect A to pin	Initial limits 25°C±5°C ambient Min	Max	Unit	Test fig
D1	1J Output short circuit current	4	-15	-100	mA	8
D2	2J	29				
D3	3J	35				
D4	4J Output short circuit current	40	-15	-100	mA	8

TABLE VII. Output leakage current test.

Test cond	Parameter	Connect A to pin	Initial limits 25°C±5°C ambient Min	Max	Unit	Test fig
E1	1J Output leakage current	4	0	1	mA	9
E2	2J	29				
E3	3J	35				
E4	4J Output leakage current	40	0	1	mA	9

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:26Z
Check the source to verify that this is the current version before use.

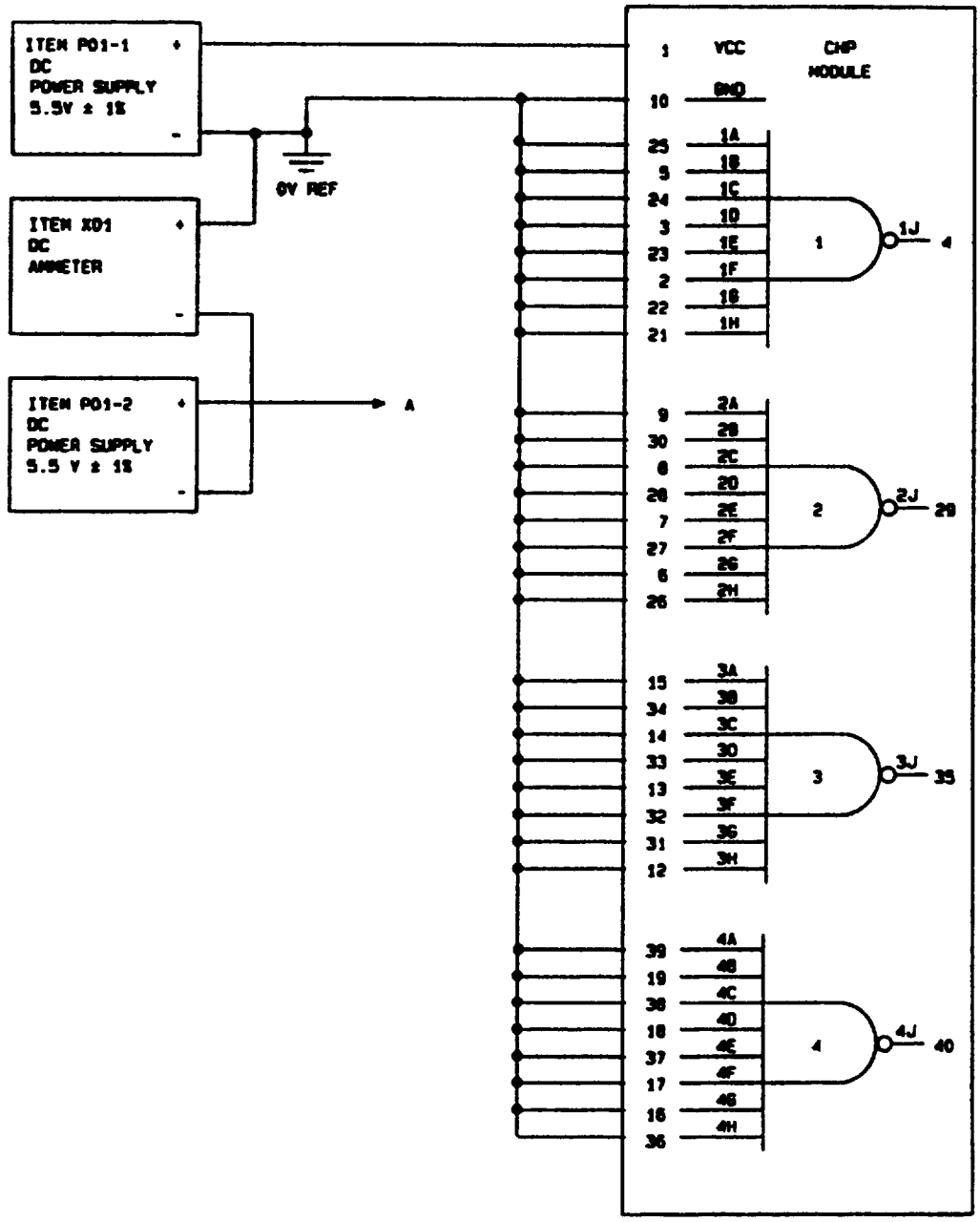


CAUTION:
Do not isolate the chassis of item X01 from earth ground.

NOTE: When performing output short circuit current test, only one output shall be shorted at a time and the duration of the test shall not exceed 1 second.

FIGURE 8. Output short circuit current test circuit for table VI.

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CAUTION:
Do not isolate the chassis of item X01 from earth ground.

NOTE: Item P01-2 shall be limited to 5 mA current.

FIGURE 9. Output leakage current test circuit for table VII.

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TABLE VIII. Input clamp diode voltage test.

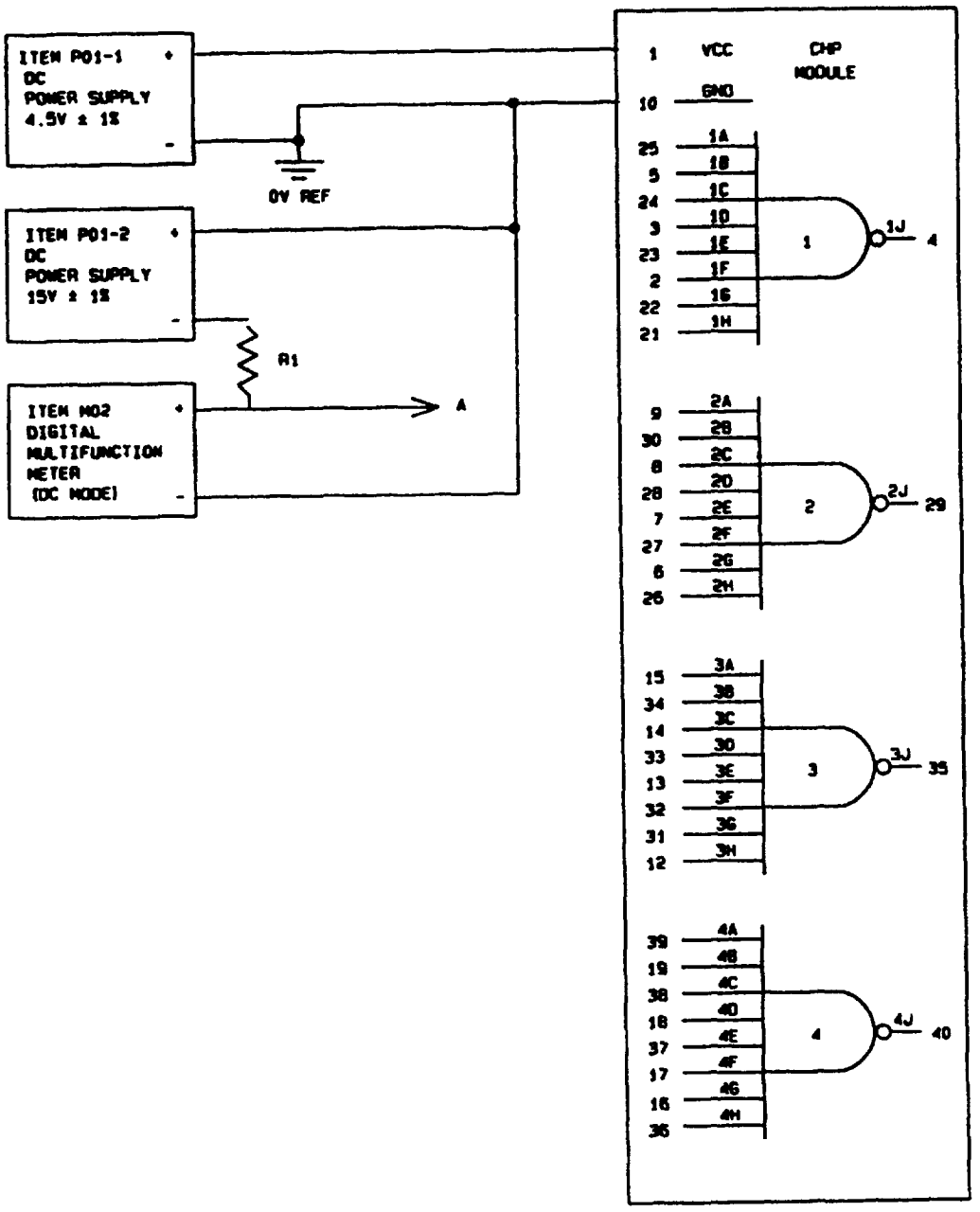
Test cond	Parameter	Connect A to pin	Initial limits 25°C±5°C ambient Min	Max	Unit	Test fig
F1	1A Input clamp diode voltage	25	-0.4	-1.5	V	10
F2	1B	5				
F3	1C	24				
F4	1D	3				
F5	1E	23				
F6	1F	2				
F7	1G	22				
F8	1H	21				
F9	2A	9				
F10	2B	30				
F11	2C	8				
F12	2D	28				
F13	2E	7				
F14	2F	27				
F15	2G	6				
F16	2H	26				
F17	3A	15				
F18	3B	34				
F19	3C	14				
F20	3D Input clamp diode voltage	33	-0.4	-1.5	V	10

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TABLE VIII. Input clamp diode voltage test - Continued.

Test cond	Parameter	Connect A to pin	Initial limits 25°C±5°C ambient Min	Max	Unit	Test fig
F21	3E Input clamp diode voltage	13	-0.4	-1.5	V	10
F22	3F	32				
F23	3G	31				
F24	3H	12				
F25	4A	39				
F26	4B	19				
F27	4C	38				
F28	4D	18				
F29	4E	37				
F30	4F	17				
F31	4G	16				
F32	4H Input clamp diode voltage	36	-0.4	-1.5	V	10

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NOTE: R1 = 750 Ohm ± 1%, 1/4 W..

*FIGURE 10. Input clamp diode voltage test circuit for table VIII.

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TABLE IX. Output voltage (low-level) test.

Test cond	Parameter	Connect			Initial limits 25°C±5°C ambient		Unit	Test fig
		A to pin	B to pin	C to pin	Min	Max		
G1	1J Output voltage, low level	25,5, 24,3, 23,2, 22,21	Open	4	0	400	mV	11
G2	2J	9,30,8, 28,7, 27,6,26		29				
G3	3J	15,34, 14,33, 13,32, 31,12		35				
G4	4J Output voltage, low level	39,19, 38,18, 18,37, 17,16, 36	Open	40	0	400	mV	11

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TABLE X. Output voltage (high-level) test.

Test cond	Parameter	Connect			Initial limits 25°C±5°C ambient		Unit	Test fig
		A to pin	B to pin	C to pin	Min	Max		
H1	1J Output voltage, high level	5,24, 3,23, 2,22, 21	25	4	2.5	4.5	V	11
H2	1J	25,24, 3,23, 2,22, 21	5					
H3	1J	25,5, 3,23, 2,22, 21	24					
H4	1J	25,5, 24,23, 2,22, 21	3					
H5	1J	25,5, 24,3, 2,22, 21	23					
H6	1J	25,5, 24,3, 23,22, 21	2					
H7	1J	25,5, 24,3, 23,2, 21	22					
H8	1J Output voltage, high level	25,5, 24,3, 23,2, 22	21	4	2.5	4.5	V	11

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TABLE X. Output voltage (high-level) test - Continued.

Test cond	Parameter	Connect			Initial limits 25°C±5°C ambient		Unit	Test fig
		A to pin	B to pin	C to pin	Min	Max		
H9	2J Output voltage, high level	30,8, 28,7, 27,6, 26	9	29	2.5	4.5	V	11
H10	2J	9,8, 28,7, 27,6, 26	30					
H11	2J	9,30, 28,7, 27,6, 26	8					
H12	2J	9,30, 8,7, 27,6, 26	28					
H13	2J	9,30, 8,28, 27,6, 26	7					
H14	2J	9,30, 8,28, 7,6,26	27					
H15	2J	9,30, 8,28, 7,27, 26	6					
H16	2J Output voltage, high level	9,30, 8,28, 7,27,6	26	29	2.5	4.5	V	11

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TABLE X. Output voltage (high-level) test - Continued.

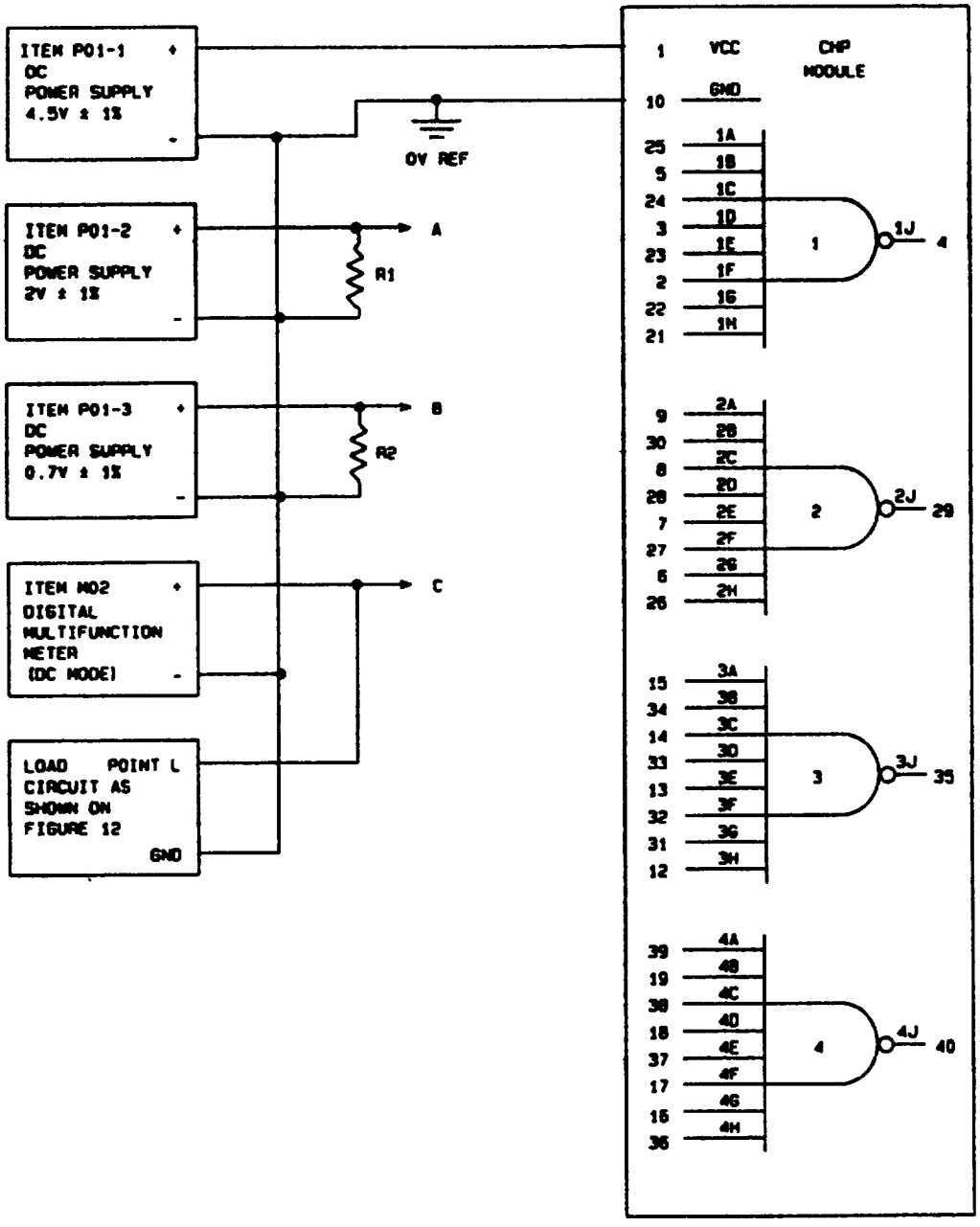
Test cond	Parameter	Connect			Initial limits 25°C±5°C ambient		Unit	Test fig
		A to pin	B to pin	C to pin	Min	Max		
H17	3J Output voltage, high level	34,14, 33,13, 32,31, 12	15	35	2.5	4.5	V	11
H18	3J	15,14, 33,13, 32,31, 12	34					
H19	3J	15,34, 33,13, 32,31, 12	14					
H20	3J	15,34, 14,13, 32,31, 12	33					
H21	3J	15,34, 14,33, 32,31, 12	13					
H22	3J	15,34, 14,33, 13,31, 12	32					
H23	3J	15,34, 14,33, 13,32, 12	31					
H24	3J Output voltage, high level	15,34, 14,33, 13,32, 31	12	35	2.5	4.5	V	11

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TABLE X. Output voltage (high-level) test - Continued.

Test cond	Parameter	Connect			Initial limits 25°C±5°C ambient		Unit	Test fig
		A to pin	B to pin	C to pin	Min	Max		
H25	4J Output voltage, high level	19,38, 18,37, 17,16, 36	39	40	2.5	4.5	V	11
H26	4J	39,38, 18,37, 17,16, 36	19					
H27	4J	39,19, 18,37, 17,16, 36	38					
H28	4J	39,19, 38,37, 17,16, 36	18					
H29	4J	39,19, 38,18, 17,16, 36	37					
H30	4J	39,19, 38,18, 37,16, 36	17					
H31	4J	39,19, 38,18, 37,17, 36	16					
H32	4J Output voltage, high level	39,19, 38,18, 37,17, 16	36	40	2.5	4.5	V	11

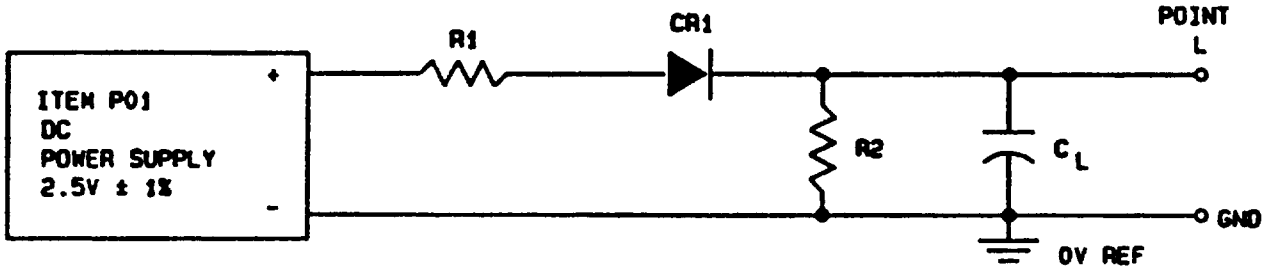
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NOTE: R1 = R2 = 51 Ohm ± 5%, 1/2 W.

FIGURE 11. Output voltage (low-level and high-level) test circuit for tables IX and X.

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NOTES:
1. CR1 = 1N3600, 1N4150, or equivalent.
2. R1 = Select value for a current flow of 4 mA ± 5% out of point L with point L held at 0.4V.
3. R2 = 6.19 kOhm ± 1%, 1/8 W.
4. Values for CL:
   (a) For test conditions M1-M4, N1-N4, P1-P32, Q1-Q32, R1-R4, and S1-S4:
   CL = 50 ± 5 pF (including probe and parasitic capacitance).
   (b) For test conditions M5-M8, N5-N8, P33-P64, and Q33-Q64:
   CL = 150 ± 5 pF (including probe and parasitic capacitance).
5. Capacitor CL is not required for dc electrical test conditions G1-G4 and H1-H32.
6. Lead length from load circuit to module output pin under test shall be less than 1 inch (25.4 mm).

FIGURE 12. CHP logic load circuit.

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TABLE XI. Input current (high-level) test.

Test cond	Parameter	Connect		Initial limits 25°C±5°C ambient		Unit	Test fig
		A to pin	B to pin	Min	Max		
J1	1A Input current, high level	2,3,5,6,7,8,9,12, 13,14,15,16,17,18, 19,21,22,23,24,26, 27,28,30,31,32,33, 34,36,37,38,39	25	0	20	µA	13
J2	1B	2,3,6,7,8,9,12,13, 14,15,16,17,18,19, 21,22,23,24,25,26, 27,28,30,31,32,33, 34,36,37,38,39	5				
J3	1C	2,3,5,6,7,8,9,12, 13,14,15,16,17,18, 19,21,22,23,25,26, 27,28,30,31,32,33 34,36,37,38,39	24				
J4	1D	2,5,6,7,8,9,12,13, 14,15,16,17,18,19, 21,22,23,24,25,26, 27,28,30,31,32,33, 34,36,37,38,39	3				
J5	1E	2,3,5,6,7,8,9,12, 13,14,15,16,17,18, 19,21,22,24,25,26, 27,28,30,31,32,33, 34,36,37,38,39	23				
J6	1F Input current, high level	3,5,6,7,8,9,12,13, 14,15,16,17,18,19, 21,22,23,24,25,26, 27,28,30,31,32,33, 34,36,37,38,39	2	0	20	µA	13

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TABLE XI. Input current (high-level) test - Continued.

Test cond		Parameter	Connect A to pin	Connect B to pin	Initial limits 25°C±5°C ambient Min	Initial limits 25°C±5°C ambient Max	Unit	Test fig
J7	1G	Input current, high level	2,3,5,6,7,8,9,12, 13,14,15,16,17,18, 19,21,23,24,25,26, 27,28,30,31,32,33, 34,36,37,38,39	22	0	20	µA	13
J8	1H		2,3,5,6,7,8,9,12, 13,14,15,16,17,18, 19,22,23,24,25,26, 27,28,30,31,32,33, 34,36,37,38,39	21				
J9	2A		2,3,5,6,7,8,12,13, 14,15,16,17,18,19, 21,22,23,24,25,26, 27,28,30,31,32,33, 34,36,37,38,39	9				
J10	2B		2,3,5,6,7,8,9,12, 13,14,15,16,17,18, 19,21,22,23,24,25, 26,27,28,31,32,33, 34,36,37,38,39	30				
J11	2C		2,3,5,6,7,9,12,13, 14,15,16,17,18,19, 21,22,23,24,25,26, 27,28,30,31,32,33, 34,36,37,38,39	8				
J12	2D	Input current, high level	2,3,5,6,7,8,9,12, 13,14,15,16,17,18, 19,21,22,23,24,25, 26,27,30,31,32,33, 34,36,37,38,39	28	0	20	µA	13

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TABLE XI. Input current (high-level) test - Continued.

Test cond	Parameter	Connect		Initial limits 25°C±5°C ambient		Unit	Test fig
		A to pin	B to pin	Min	Max		
J13	2E Input current, high level	2,3,5,6,8,9,12,13, 14,15,16,17,18,19, 21,22,23,24,25,26, 27,28,30,31,32,33, 34,36,37,38,39	7	0	20	µA	13
J14	2F	2,3,5,6,7,8,9,12, 13,14,15,16,17,18, 19,21,22,23,24,25, 26,28,30,31,32,33, 34,36,37,38,39	27				
J15	2G	2,3,5,7,8,9,12,13, 14,15,16,17,18,19, 21,22,23,24,25,26, 27,28,30,31,32,33, 34,36,37,38,39	6				
J16	2H	2,3,5,6,7,8,9,12, 13,14,15,16,17,18, 19,21,22,23,24,25, 27,28,30,31,32,33, 34,36,37,38,39	26				
J17	3A	2,3,5,6,7,8,9,12, 13,14,16,17,18,19, 21,22,23,24,25,26, 27,28,30,31,32,33, 34,36,37,38,39	15				
J18	3B Input current, high level	2,3,5,6,7,8,9,12, 13,14,15,16,17,18, 19,21,22,23,24,25, 26,27,28,30,31,32, 33,36,37,38,39	34	0	20	µA	13

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T17:26Z
Check the source to verify that this is the current version before use.

TABLE XI. Input current (high-level) test - Continued.

Test cond	Parameter	Connect		Initial limits 25°C±5°C ambient		Unit	Test fig
		A to pin	B to pin	Min	Max		
J19	3C Input current, high level	2,3,5,6,7,8,9,12, 13,15,16,17,18,19, 21,22,23,24,25,26, 27,28,30,31,32,33, 34,36,37,38,39	14	0	20	µA	13
J20	3D	2,3,5,6,7,8,9,12, 13,14,15,16,17,18, 19,21,22,23,24,25, 26,27,28,30,31,32, 34,36,37,38,39	33				
J21	3E	2,3,5,6,7,8,9,12, 14,15,16,17,18,19, 21,22,23,24,25,26, 27,28,30,31,32,33, 34,36,37,38,39	13				
J22	3F	2,3,5,6,7,8,9,12, 13,14,15,16,17,18, 19,21,22,23,24,25, 26,27,28,30,31,33, 34,36,37,38,39	32				
J23	3G	2,3,5,6,7,8,9,12, 13,14,15,16,17,18, 19,21,22,23,24,25, 26,27,28,30,32,33, 34,36,37,38,39	31				
J24	3H Input current, high level	2,3,5,6,7,8,9,13, 14,15,16,17,18,19, 21,22,23,24,25,26, 27,28,30,31,32,33, 34,36,37,38,39	12	0	20	µA	13

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TABLE XI. Input current (high-level) test - Continued.

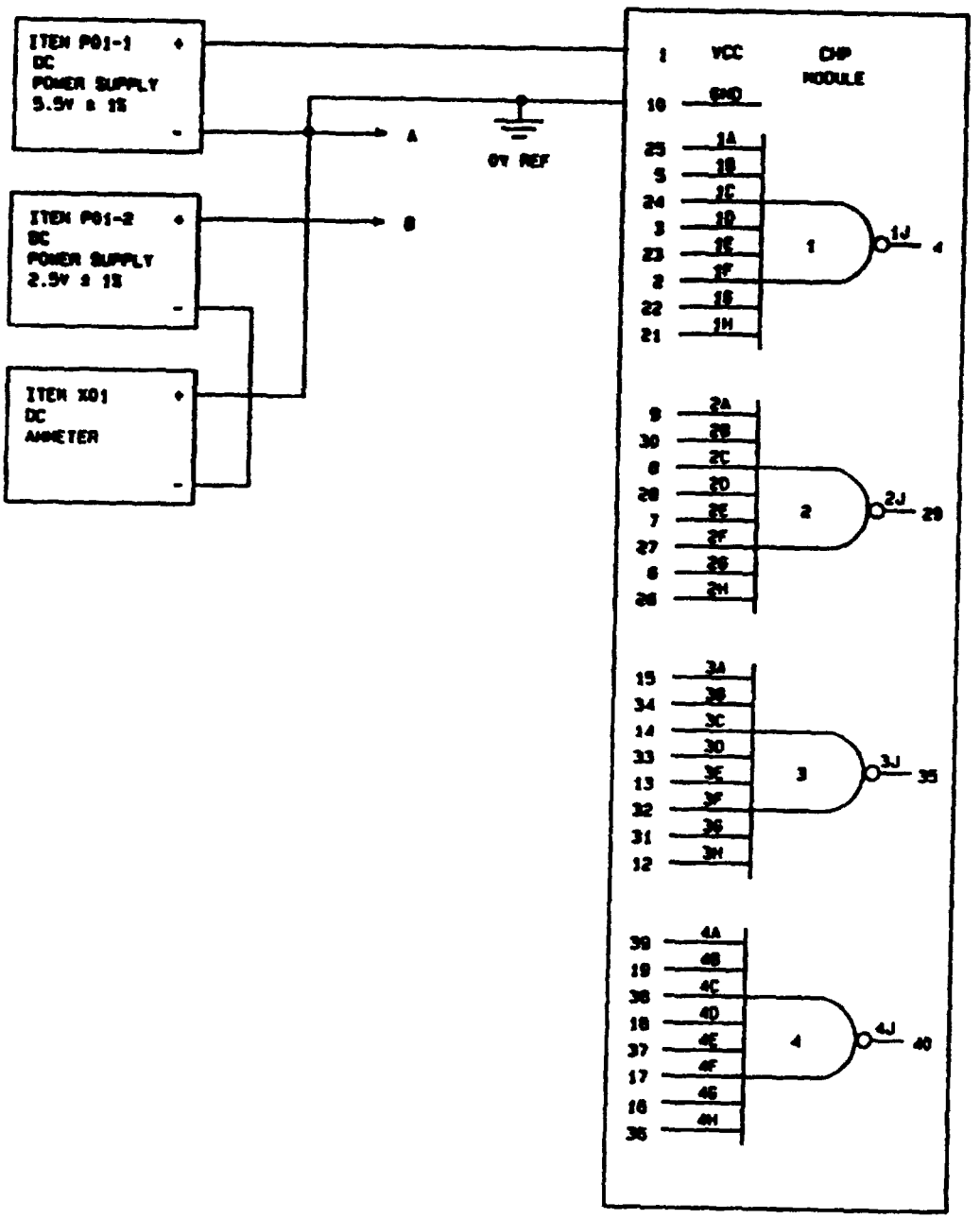
Test cond		Parameter	Connect A to pin	Connect B to pin	Initial limits 25°C±5°C ambient Min	Initial limits 25°C±5°C ambient Max	Unit	Test fig
J25	4A	Input current, high level	2,3,5,6,7,8,9,12, 13,14,15,16,17,18, 19,21,22,23,24,25, 26,27,28,30,31,32, 33,34,36,37,38	39	0	20	µA	13
J26	4B		2,3,5,6,7,8,9,12, 13,14,15,16,17,18, 21,22,23,24,25,26, 27,28,30,31,32,33, 34,36,37,38,39	19				
J27	4C		2,3,5,6,7,8,9,12, 13,14,15,16,17,18, 19,21,22,23,24,25, 26,27,28,30,31,32, 33,34,36,37,39	38				
J28	4D		2,3,5,6,7,8,9,12, 13,14,15,16,17,19, 21,22,23,24,25,26, 27,28,30,31,32,33, 34,36,37,38,39	18				
J29	4E		2,3,5,6,7,8,9,12, 13,14,15,16,17,18, 19,21,22,23,24,25, 26,27,28,30,31,32, 33,34,36,38,39	37				
J30	4F	Input current, high level	2,3,5,6,7,8,9,12, 13,14,15,16,18,19, 21,22,23,24,25,26, 27,28,30,31,32,33, 34,36,37,38,39	17	0	20	µA	13

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TABLE XI. Input current (high-level) test - Continued.

Test cond	Parameter	Connect		Initial limits 25°C±5°C ambient		Unit	Test fig
		A to pin	B to pin	Min	Max		
J31	4G Input current, high level	2,3,5,6,7,8,9,12, 13,14,15,17,18,19, 21,22,23,24,25,26, 27,28,30,31,32,33, 34,36,37,38,39	16	0	20	µA	13
J32	4H Input current, high level	2,3,5,6,7,8,9,12, 13,14,15,16,17,18, 19,21,22,23,24,25, 26,27,28,30,31,32, 33,34,37,38,39	36	0	20	µA	13

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CAUTION:
Do not isolate the chassis of item X01 from earth ground.

FIGURE 13. Input current (high-level) test circuit for table XI.

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TABLE XII. Input current (low-level) test.

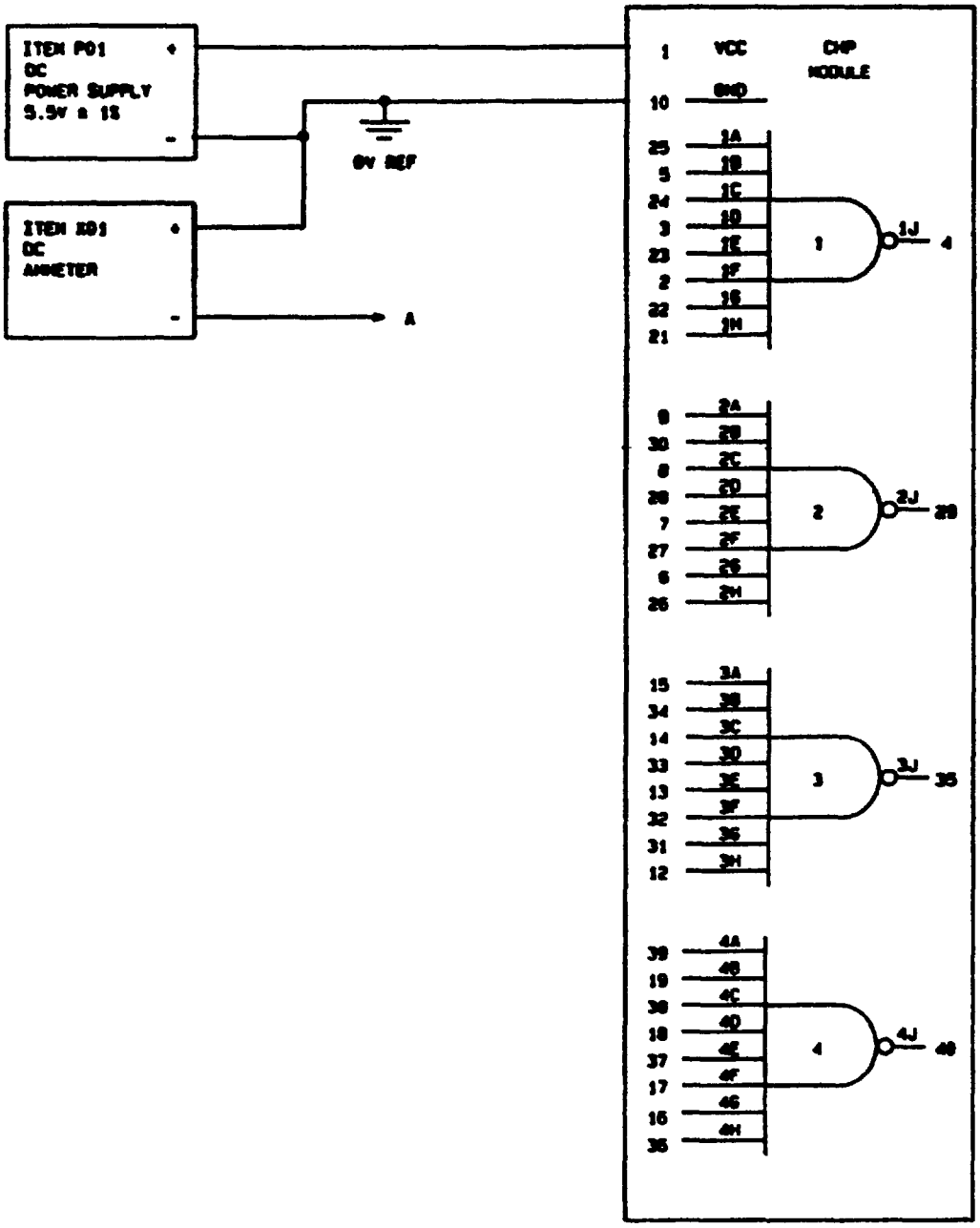
Test cond	Parameter	Connect A to pin	Initial limits 25°C±5°C ambient Min	Initial limits 25°C±5°C ambient Max	Unit	Test fig
K1	1A Input current, low level	25	-10	-434	µA	14
K2	1B	5				
K3	1C	24				
K4	1D	3				
K5	1E	23				
K6	1F	2				
K7	1G	22				
K8	1H	21				
K9	2A	9				
K10	2B	30				
K11	2C	8				
K12	2D	28				
K13	2E	7				
K14	2F	27				
K15	2G	6				
K16	2H	26				
K17	3A	15				
K18	3B	34				
K19	3C	14				
K20	3D Input current, low level	33	-10	-434	µA	14

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TABLE XII. Input current (low-level) test - Continued.

Test cond	Parameter	Connect A to pin	Initial limits 25°C±5°C ambient Min	Initial limits 25°C±5°C ambient Max	Unit	Test fig
K21	3E Input current, low level	13	-10	-434	µA	14
K22	3F	32				
K23	3G	31				
K24	3H	12				
K25	4A	39				
K26	4B	19				
K27	4C	38				
K28	4D	18				
K29	4E	37				
K30	4F	17				
K31	4G	16				
K32	4H Input current, low level	36	-10	-434	µA	14

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CAUTION:
Do not isolate the chassis of item X01 from earth ground.

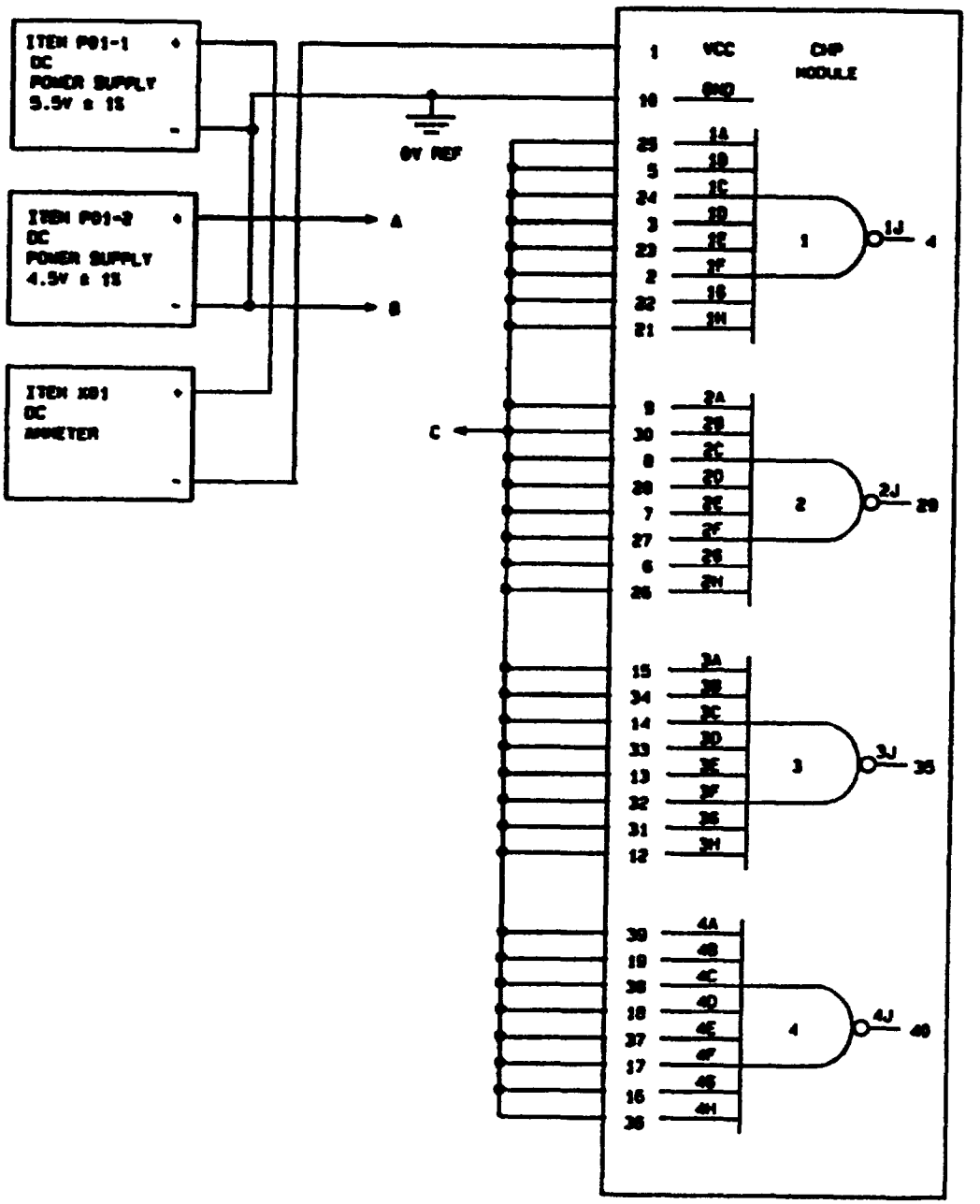
FIGURE 14. Input current (low-level) test circuit for table XII.

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TABLE XIII. Power supply current (low-level and high-level outputs) test.

Test cond	Parameter	Connect		Initial limits 25°C±5°C ambient		Unit	Test fig
		A to pin	B to pin	Min	Max		
L1	Power supply current, low level outputs	C	Open	0.1	4.4	mA	15
L2	Power supply current, high level outputs	Open	C	0.1	2.0	mA	15

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CAUTION:
Do not isolate the chassis of item X01 from earth ground.

FIGURE 15. Power supply current (low-level and high-level outputs) test circuit for table XIII.

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*TABLE XIV. Transition time (tTLH and tTHL) and propagation delay time (tPLH and tPHL) test.

Test cond	Parameter	Connect			Initial limits 25±5°C ambient		Unit	Test fig
		A to pin	B to pin	C to pin	Min	Max		
M1	1J Transition time (tTLH), CL = 50 pF	5,24, 3,23, 2,22, 21	25	4	5	50	ns	16,17
M2	2J	30,8, 28,7, 27,6, 26	9	29				
M3	3J	34, 14, 33, 13, 32, 31,12	15	35				
M4	4J Transition time (tTLH), CL = 50 pF	19, 38, 18, 37, 17, 16,36	39	40	5	50		
M5	1J Transition time (tTLH), CL = 150 pF	5,24, 3,23, 2,22, 21	25	4	10	80		
M6	2J	30,8, 28,7, 27,6, 26	9	29				
M7	3J Transition time (tTLH), CL = 150 pF	34, 14, 33, 13, 32, 31,12	15	35	10	80	ns	16,17

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*TABLE XIV. Transition time (tTLH and tTHL) and propagation delay time (tPLH and tPHL) test - Continued.

Test cond	Parameter	Connect			Initial limits 25±5°C ambient		Unit	Test fig
		A to pin	B to pin	C to pin	Min	Max		
M8	4J Transition time (tTLH), CL = 150 pF	19, 38, 18, 37, 17, 16,36	39	40	10	80	ns	16,17
N1	1J Transition time (tTHL), CL = 50 pF	5,24, 3,23, 2,22, 21	25	4	3	18		
N2	2J	30,8, 28,7, 27,6, 26	9	29				
N3	3J	34, 14, 33, 13, 32, 31,12	15	35				
N4	4J Transition time (tTHL), CL = 50 pF	19, 38, 18, 37, 17, 16,36	39	40	3	18		
N5	1J Transition time (tTHL), CL = 150 pF	25, 24,3, 23,2, 22,21	25	4	6	40	ns	16,17

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*TABLE XIV. Transition time (tTLH and tTHL) and propagation delay time (tPLH and tPHL) test - Continued.

Test cond	Parameter	Connect			Initial limits 25±5°C ambient		Unit	Test fig
		A to pin	B to pin	C to pin	Min	Max		
N6	2J Transition time (tTHL), CL = 150 pF	30,8, 28,7, 27,6, 26	9	29	6	40	ns	16,17
N7	3J	34, 14, 33, 13, 32, 31,12	15	35	6	40		
N8	4J Transition time (tTHL), CL = 150 pF	19, 38, 18, 37, 17, 16,36	39	40	6	40		
P1	1A to 1J Propagation delay time (tPLH), CL = 50 pF	5,24, 3,23, 2,22, 21	25	4	3	25		
P2	1B to 1J	25, 24,3, 23,2, 22,21	5					
P3	1C to 1J	25,5, 3,23, 2,22, 21	24					
P4	1D to 1J Propagation delay time (tPLH), CL = 50 pF	25,5, 24, 23,2, 22,21	3	4	3	25	ns	16,17

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*TABLE XIV. Transition time (tTLH and tTHL) and propagation delay time (tPLH and tPHL) test - Continued.

Test cond	Parameter	Connect			Initial limits 25±5°C ambient		Unit	Test fig
		A to pin	B to pin	C to pin	Min	Max		
P5	1E to 1J Propagation delay time (tPLH), CL = 50 pF	25,5, 24,3, 2,22, 21	23	4	3	25	ns	16,17
P6	1F to 1J	25,5, 24,3, 23, 22,21	2					
P7	1G to 1J	25,5, 24,3, 23,2, 21	22	4				
P8	1H to 1J	25,5, 24,3, 23,2, 22	21	40				
P9	2A to 2J	30,8, 28,7, 27,6, 26	9	29				
P10	2B to 2J	9,8, 28,7, 27,6, 26	30					
P11	2C to 2J	9,30, 28,7, 27,6, 26	8					
P12	2D to 2J Propagation delay time (tPLH), CL = 50 pF	9,30, 8,7, 27,6, 26	28	29	3	25	ns	16,17

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*TABLE XIV. Transition time (tTLH and tTHL) and propagation delay time (tPLH and tPHL) test - Continued.

Test cond	Parameter	Connect			Initial limits 25±5°C ambient		Unit	Test fig
		A to pin	B to pin	C to pin	Min	Max		
P13	2E to 2J Propagation delay time (tPLH), CL = 50 pF	9,30, 8,28, 27,6, 26	7	29	3	25	ns	16,17
P14	2F to 2J	9,30, 8,28, 7,6, 26	27					
P15	2G to 2J	9,30, 8,28, 7,27, 26	6					
P16	2H to 2J	9,30, 8,28, 7,27, 6	26	29				
P17	3A to 3J	34, 14, 33, 13, 32, 31,12	15	35				
P18	3B to 3J Propagation delay time (tPLH), CL = 50 pF	15, 14, 33, 13, 32, 31,12	34	35	3	25	ns	16,17

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*TABLE XIV. Transition time (tTLH and tTHL) and propagation delay time (tPLH and tPHL) test - Continued.

Test cond	Parameter	Connect			Initial limits 25±5°C ambient		Unit	Test fig
		A to pin	B to pin	C to pin	Min	Max		
P19	3C to 3J Propagation delay time (tPLH), CL = 50 pF	15, 34, 33, 13, 32, 31,12	14	35	3	25	ns	16,17
P20	3D to 3J	15, 34, 14, 13, 32, 31,12	33					
P21	3E to 3J	15, 34, 14, 33, 32, 31,12	13					
P22	3F to 3J	15, 34, 14, 33, 13, 31,12	32					
P23	3G to 3J Propagation delay time (tPLH), CL = 50 pF	15, 34, 14, 33, 13, 32,12	31	35	3	25	ns	16,17

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*TABLE XIV. Transition time (tTLH and tTHL) and propagation delay time (tPLH and tPHL) test - Continued.

Test cond	Parameter	Connect			Initial limits 25±5°C ambient		Unit	Test fig
		A to pin	B to pin	C to pin	Min	Max		
P24	3H to 3J Propagation delay time (tPLH), CL = 50 pF	15, 34, 14, 33, 13, 32,31	12	35	3	25	ns	16,17
P25	4A to 4J	19, 38, 18, 37, 17, 16,36	39	40				
P26	4B to 4J	39, 38, 18, 37, 17, 16,36	19					
P27	4C to 4J	39, 19, 18, 37, 17, 16,36	38					
P28	4D to 4J Propagation delay time (tPLH), CL = 50 pF	39, 19, 38, 37, 17, 16,36	18	40	3	25	ns	16,17

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*TABLE XIV. Transition time (tTLH and tTHL) and propagation delay time (tPLH and tPHL) test - Continued.

Test cond	Parameter	Connect			Initial limits 25±5°C ambient		Unit	Test fig
		A to pin	B to pin	C to pin	Min	Max		
P29	4E to 4J Propagation delay time (tPLH), CL = 50 pF	39, 19, 38, 18, 17, 16,36	37	40	3	25	ns	16,17
P30	4F to 4J	39, 19, 38, 18, 37, 16,36	17					
P31	4G to 4J	39, 19, 38, 18, 37, 17,36	16					
P32	4H to 4J Propagation delay time (tPLH), CL = 50 pF	39, 19, 38, 18, 37, 17,16	36	40	3	25		
P33	1A to 1J Propagation delay time (tPLH), CL = 150 pF	5,24, 3,23, 2,22, 21	25	4	4	35		
P34	1B to 1J Propagation delay time (tPLH), CL = 150 pF	25, 24,3, 23,2, 22,21	5	4	4	35	ns	16,17

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*TABLE XIV. Transition time (tTLH and tTHL) and propagation delay time (tPLH and tPHL) test - Continued.

Test cond	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25±5°C ambient Min	Initial limits 25±5°C ambient Max	Unit	Test fig
P35	1C to 1J Propagation delay time (tPLH), CL = 150 pF	25,5, 3,23, 2,22, 21	24	4	4	35	ns	16,17
P36	1D to 1J	25,5, 24, 23,2, 22,21	3					
P37	1E to 1J	25,5, 24,3, 2,22, 21	23					
P38	1F to 1J	25,5, 24,3, 23, 22,21	2					
P39	1G to 1J	25,5, 24,3, 23,2, 21	22					
P40	1H to 1J	25,5, 24,3, 23,2, 22	21	4				
P41	2A to 2J	30,8, 28,7, 27,6, 26	9	29				
P42	2B to 2J Propagation delay time (tPLH), CL = 150 pF	9,8, 28,7, 27,6, 26	30	29	4	35	ns	16,17

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*TABLE XIV. Transition time (tTLH and tTHL) and propagation delay time (tPLH and tPHL) test - Continued.

Test cond	Parameter	Connect			Initial limits 25±5°C ambient		Unit	Test fig
		A to pin	B to pin	C to pin	Min	Max		
P43	2C to 2J Propagation delay time (tPLH), CL = 150 pF	9,30, 28,7, 27,6, 26	8	29	4	35	ns	16,17
P44	2D to 2J	9,30, 8,7, 27,6, 26	28					
P45	2E to 2J	9,30, 8,28, 27,6, 26	7					
P46	2F to 2J	9,30, 8,28, 7,6, 26	27					
P47	2G to 2J	9,30, 8,28, 7,27, 26	6					
P48	2H to 2J	9,30, 8,28, 7,27, 6	26	29				
P49	3A to 3J Propagation delay time (tPLH), CL = 150 pF	34, 14, 33, 13, 32, 31,12	15	35	4	35	ns	16,17

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*TABLE XIV. Transition time (tTLH and tTHL) and propagation delay time (tPLH and tPHL) test - Continued.

Test cond	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25±5°C ambient Min	Initial limits 25±5°C ambient Max	Unit	Test fig
P50	3B to 3J Propagation delay time (tPLH), CL = 150 pF	15, 14, 33, 13, 32, 31,12	34	35	4	35	ns	16,17
P51	3C to 3J	15, 34, 33, 13, 32, 31,12	14					
P52	3D to 3J	15, 34, 14, 13, 32, 31,12	33					
P53	3E to 3J	15, 34, 14, 33, 32, 31,12	13					
P54	3F to 3J Propagation delay time (tPLH), CL = 150 pF	15, 34, 14, 33, 13, 31,12	32	35	4	35	ns	16,17

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*TABLE XIV. Transition time (tTLH and tTHL) and propagation delay time (tPLH and tPHL) test - Continued.

Test cond	Parameter	Connect			Initial limits 25±5°C ambient		Unit	Test fig
		A to pin	B to pin	C to pin	Min	Max		
P55	3G to 3J Propagation delay time (tPLH), CL = 150 pF	15, 34, 14, 33, 13, 32,12	31	35	4	35	ns	16,17
P56	3H to 3J	15, 34, 14, 33, 13, 32, 31	12	35				
P57	4A to 4J	19, 38, 18, 37, 17, 16,36	39	40				
P58	4B to 4J	39, 38, 18, 37, 17, 16,36	19					
P59	4C to 4J Propagation delay time (tPLH), CL = 150 pF	39, 19, 18, 37, 17, 16,36	38	40	4	35	ns	16,17

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*TABLE XIV. Transition time (tTLH and tTHL) and propagation delay time (tPLH and tPHL) test - Continued.

Test cond	Parameter	Connect			Initial limits 25±5°C ambient		Unit	Test fig
		A to pin	B to pin	C to pin	Min	Max		
P60	4D to 4J Propagation delay time (tPLH), CL = 150 pF	39, 19, 38, 37, 17, 16,36	18	40	4	35	ns	16,17
P61	4E to 4J	39, 19, 38, 18, 17, 16,36	37					
P62	4F to 4J	39, 19, 38, 18, 37, 16,36	17					
P63	4G to 4J	39, 19, 38, 18, 37, 17,36	16					
P64	4H to 4J Propagation delay time (tPLH), CL = 150 pF	39, 19, 38, 18, 37, 17,16	36	40	4	35	ns	16,17

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*TABLE XIV. Transition time (tTLH and tTHL) and propagation delay time (tPLH and tPHL) test - Continued.

Test cond	Parameter	Connect			Initial limits 25±5°C ambient		Unit	Test fig
		A to pin	B to pin	C to pin	Min	Max		
Q1	1A to 1J Propagation delay time (tPHL), CL = 50 pF	5,24, 3,23, 2,22, 21	25	4	3	35	ns	16,17
Q2	1B to 1J	25, 24,3, 23,2, 22,21	5					
Q3	1C to 1J	25, 5,3, 23,2, 22,21	24					
Q4	1D to 1J	25,5, 24, 23,2, 22,21	3					
Q5	1E to 1J	25,5, 24,3, 2,22, 21	23					
Q6	1F to 1J	25,5, 24,3, 23, 22,21	2					
Q7	1G to 1J Propagation delay time (tPHL), CL = 50 pF	25,5, 24,3, 23,2, 21	22	4	3	35	ns	16,17

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*TABLE XIV. Transition time (tTLH and tTHL) and propagation delay time (tPLH and tPHL) test - Continued.

Test cond	Parameter	Connect			Initial limits 25±5°C ambient		Unit	Test fig
		A to pin	B to pin	C to pin	Min	Max		
Q8	1H to 1J Propagation delay time (tPHL), CL = 50 pF	25,5, 24,3, 23,2, 22	21	4	3	35	ns	16,17
Q9	2A to 2J	30,8, 28,7, 27,6, 26	9	29				
Q10	2B to 2J	9,8, 28,7, 27,6, 26	30					
Q11	2C to 2J	9,30, 28,7, 27,6, 26	8					
Q12	2D to 2J	9,30, 8,7, 27,6, 26	28					
Q13	2E to 2J	9,30, 8,28, 27,6, 26	7					
Q14	2F to 2J Propagation delay time (tPHL), CL = 50 pF	9,30, 8,28, 7,6, 26	27	29	3	35	ns	16,17

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*TABLE XIV. Transition time (tTLH and tTHL) and propagation delay time (tPLH and tPHL) test - Continued.

Test cond	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25±5°C ambient Min	Initial limits 25±5°C ambient Max	Unit	Test fig
Q15	2G to 2J Propagation delay time (tPHL), CL = 50 pF	9,30, 8,28, 7,27, 26	6	29	3	35	ns	16,17
Q16	2H to 2J	9,30, 8,28, 7,27, 6	26	29				
Q17	3A to 3J	34, 14, 33, 13, 32, 31,12	15	35				
Q18	3B to 3J	15, 14, 33, 13, 32, 31,12	34					
Q19	3C to 3J Propagation delay time (tPHL), CL = 50 pF	15, 34, 33, 13, 32, 31,12	14	35	3	35	ns	16,17

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*TABLE XIV. Transition time (tTLH and tTHL) and propagation delay time (tPLH and tPHL) test - Continued.

Test cond	Parameter	Connect			Initial limits 25±5°C ambient		Unit	Test fig
		A to pin	B to pin	C to pin	Min	Max		
Q20	3D to 3J Propagation delay time (tPHL), CL = 50 pF	15, 34, 14, 13, 32, 31,12	33	35	3	35	ns	16,17
Q21	3E to 3J	15, 34, 14, 33, 32, 31,12	13					
Q22	3F to 3J	15, 34, 14, 33, 13, 31,12	32					
Q23	3G to 3J	15, 34, 14, 33, 13, 32,12	31					
Q24	3H to 3J Propagation delay time (tPHL), CL = 50 pF	15, 34, 14, 33, 13, 32,31	12	35	3	35	ns	16,17

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*TABLE XIV. Transition time (tTLH and tTHL) and propagation delay time (tPLH and tPHL) test - Continued.

Test cond	Parameter	Connect			Initial limits 25±5°C ambient		Unit	Test fig
		A to pin	B to pin	C to pin	Min	Max		
Q25	4A to 4J Propagation delay time (tPHL), CL = 50 pF	19, 38, 18, 37, 17, 16,36	39	40	3	35	ns	16,17
Q26	4B to 4J	39, 38, 18, 37, 17, 16,36	19					
Q27	4C to 4J	39, 19, 18, 37, 17, 16,36	38					
Q28	4D to 4J	39, 19, 38, 37, 17, 16,36	18					
Q29	4E to 4J Propagation delay time (tPHL), CL = 50 pF	39, 19, 38, 18, 17, 16,36	37	40	3	35	ns	16,17

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*TABLE XIV. Transition time (tTLH and tTHL) and propagation delay time (tPLH and tPHL) test - Continued.

Test cond	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Initial limits 25±5°C ambient Min	Initial limits 25±5°C ambient Max	Unit	Test fig
Q30	4F to 4J Propagation delay time (tPHL), CL = 50 pF	39, 19, 38, 18, 37, 16,36	17	40	3	35	ns	16,17
Q31	4G to 4J	39, 19, 38, 18, 37, 17,36	16					
Q32	4H to 4J Propagation delay time (tPHL), CL = 50 pF	39, 19, 38, 18, 37, 17,16	36	40	3	35		
Q33	1A to 1J Propagation delay time (tPHL), CL = 150 pF	5,24, 3,23, 2,22, 21	25	4	4	45		
Q34	1B to 1J	25, 24,3, 23,2, 22,21	5					
Q35	1C to 1J Propagation delay time (tPHL), CL = 150 pF	25,5, 3,23, 2,22, 21	24	4	4	45	ns	16,17

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*TABLE XIV. Transition time (tTLH and tTHL) and propagation delay time (tPLH and tPHL) test - Continued.

Test cond	Parameter	Connect			Initial limits 25±5°C ambient		Unit	Test fig
		A to pin	B to pin	C to pin	Min	Max		
Q36	1D to 1J Propagation delay time (tPHL), CL = 150 pF	25,5, 24, 23,2, 22,21	3	4	4	45	ns	16,17
Q37	1E to 1J	25,5, 24,3, 2,22, 21	23					
Q38	1F to 1J	25,5, 24,3, 23, 22,21	2					
Q39	1G to 1J	25,5, 24,3, 23,2, 21	22					
Q40	1H to 1J	25,5, 24,3, 23,2, 22	21	4				
Q41	2A to 2J	30,8, 28,7, 27,6, 26	9	29				
Q42	2B to 2J Propagation delay time (tPHL), CL = 150 pF	9,8, 28,7, 27,6, 26	30	29	4	45	ns	16,17

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*TABLE XIV. Transition time (tTLH and tTHL) and propagation delay time (tPLH and tPHL) test - Continued.

Test cond	Parameter	Connect			Initial limits 25±5°C ambient		Unit	Test fig
		A to pin	B to pin	C to pin	Min	Max		
Q43	2C to 2J Propagation delay time (tPHL), CL = 150 pF	9,30, 28,7, 27,6, 26	8	29	4	45	ns	16,17
Q44	2D to 2J	9,30, 8,7, 27,6, 26	28					
Q45	2E to 2J	9,30, 8,28, 27,6, 26	7					
Q46	2F to 2J	9,30, 8,28, 7,6, 26	27					
Q47	2G to 2J	9,30, 8,28, 7,27, 26	6					
Q48	2H to 2J	9,30, 8,28, 7,27, 26	26	29				
Q49	3A to 3J Propagation delay time (tPHL), CL = 150 pF	34, 14, 33, 13, 32, 31,12	15	35	4	45	ns	16,17

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*TABLE XIV. Transition time (tTLH and tTHL) and propagation delay time (tPLH and tPHL) test - Continued.

Test cond	Parameter	Connect			Initial limits 25±5°C ambient		Unit	Test fig
		A to pin	B to pin	C to pin	Min	Max		
Q50	3B to 3J Propagation delay time (tPHL), CL = 150 pF	15, 14, 33, 13, 32, 31,12	34	35	4	45	ns	16,17
Q51	3C to 3J	15, 34, 33, 13, 32, 31,12	14					
Q52	3D to 3J	15, 34, 14, 13, 32, 31,12	33					
Q53	3E to 3J	15, 34, 14, 33, 32, 31,12	13					
Q54	3F to 3J Propagation delay time (tPHL), CL = 150 pF	15, 34, 14, 33, 13, 31,12	32	35	4	45	ns	16,17

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*TABLE XIV. Transition time (tTLH and tTHL) and propagation delay time (tPLH and tPHL) test - Continued.

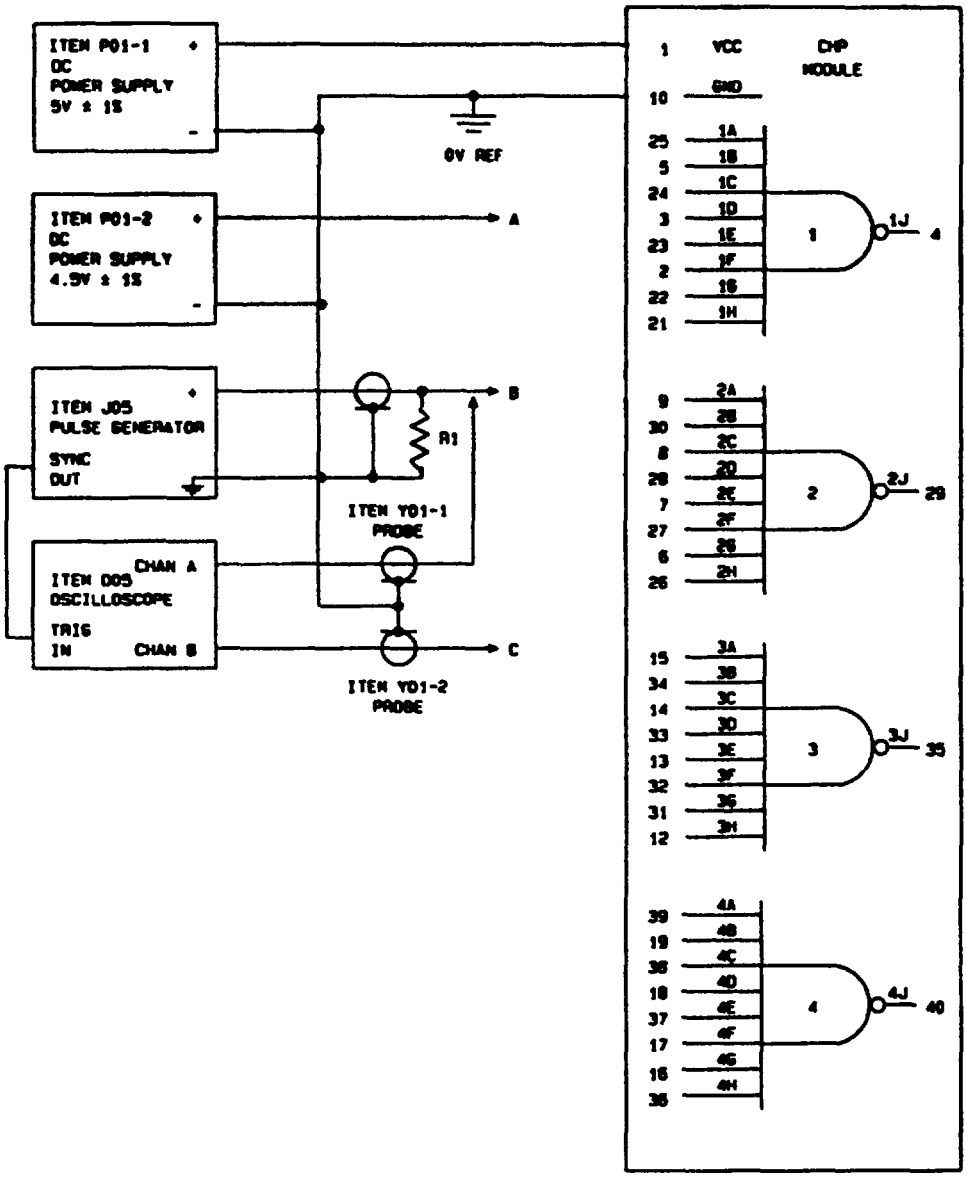
Test cond	Parameter	Connect			Initial limits 25±5°C ambient		Unit	Test fig
		A to pin	B to pin	C to pin	Min	Max		
Q55	3G to 3J Propagation delay time (tPHL) CL = 150 pF	15, 34, 14, 33, 13, 32,12	31	35	4	45	ns	16,17
Q56	3H to 3J	15, 34, 14, 33, 13, 32,31	12	35				
Q57	4A to 4J	19, 38, 18, 37, 17, 16,36	39	40				
Q58	4B to 4J	39, 38, 18, 37, 17, 16,36	19					
Q59	4C to 4J Propagation delay time (tPHL), CL = 150 pF	39, 19, 18, 37, 17, 16,36	38	40	4	45	ns	16,17

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*TABLE XIV. Transition time (tTLH and tTHL) and propagation delay time (tPLH and tPHL) test - Continued.

Test cond	Parameter	Connect			Initial limits 25±5°C ambient		Unit	Test fig
		A to pin	B to pin	C to pin	Min	Max		
Q60	4D to 4J Propagation delay time (tPHL), CL = 150 pF	39, 19, 38, 37, 17, 16,36	18	40	4	45	ns	16,17
Q61	4E to 4J	39, 19, 38, 18, 17, 16,36	37					
Q62	4F to 4J	39, 19, 38, 18, 37, 16,36	17					
Q63	4G to 4J	39, 19, 38, 18, 37, 17,36	16					
Q64	4H to 4J Propagation delay time (tPHL), CL = 150 pF	39, 19, 38, 18, 37, 17,16	36	40	4	45	ns	16,17

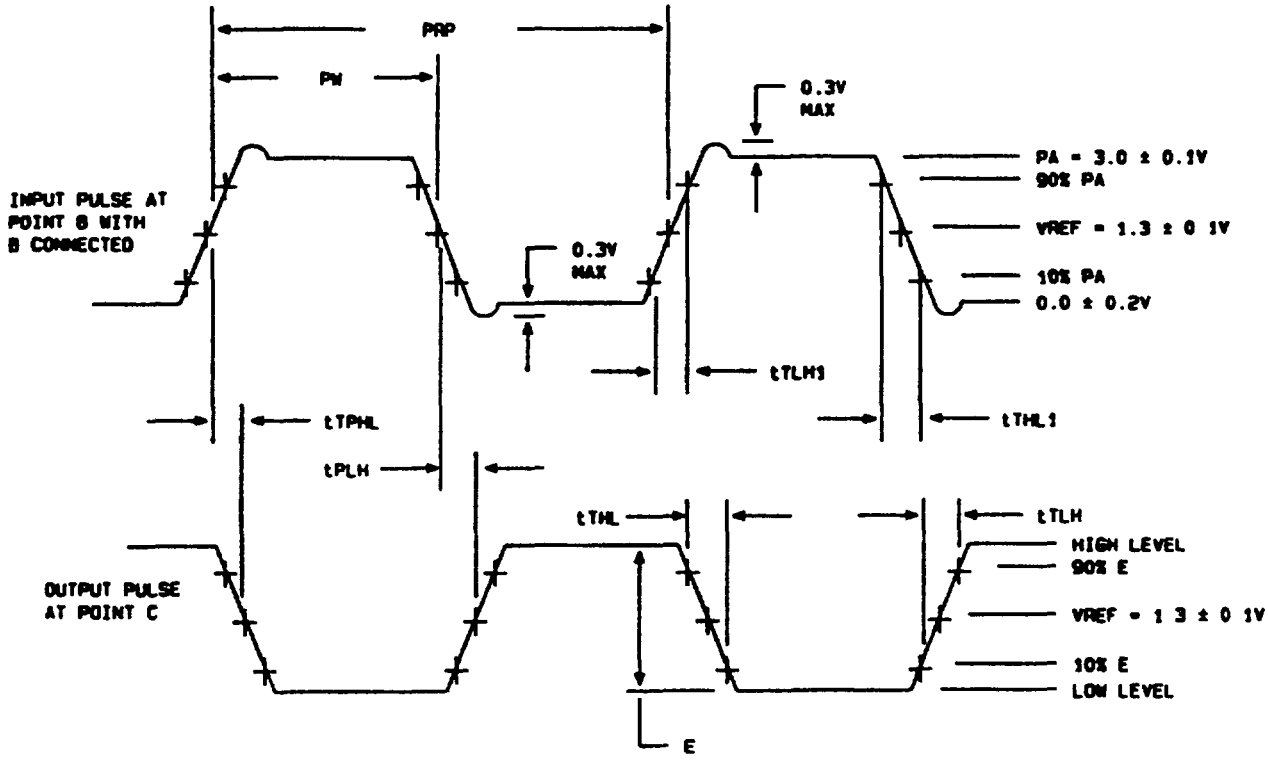
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NOTES:
1. R1 = 51 Ohm ± 5%, 1/2 W.
2. Load output with load as specified on figure 12.

FIGURE 16. Transition time (tTLH and tTHL) and propagation delay time (tPLH and tPHL) test circuit for table XIV.

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NOTES:

1. Input pulse parameters:
   tTLH1 = tTHL1 = 6 ± 1 ns.
   PW = 500 ± 50 ns.
   PRP = 1000 ± 100 ns.

2. Before making the measurements, verify that the output pulse high level is $\geq$ 2.5 V and the low level is $\leq$ 0.4 V.

3. The lead length of electrical conductors between resistor R1 and input pins; between items Y01-1 and Y01-2 and input and output pins; and between load circuits and output pins shall be less than 1 inch (25.4 mm).

4. All voltages are referenced to ground.

FIGURE 17. Transition time (tTLH and tTHL) and propagation delay time (tPLH and tPHL) test circuit waveforms for table XIV.

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TABLE XV. Output crosstalk (low-level and high-level) test.

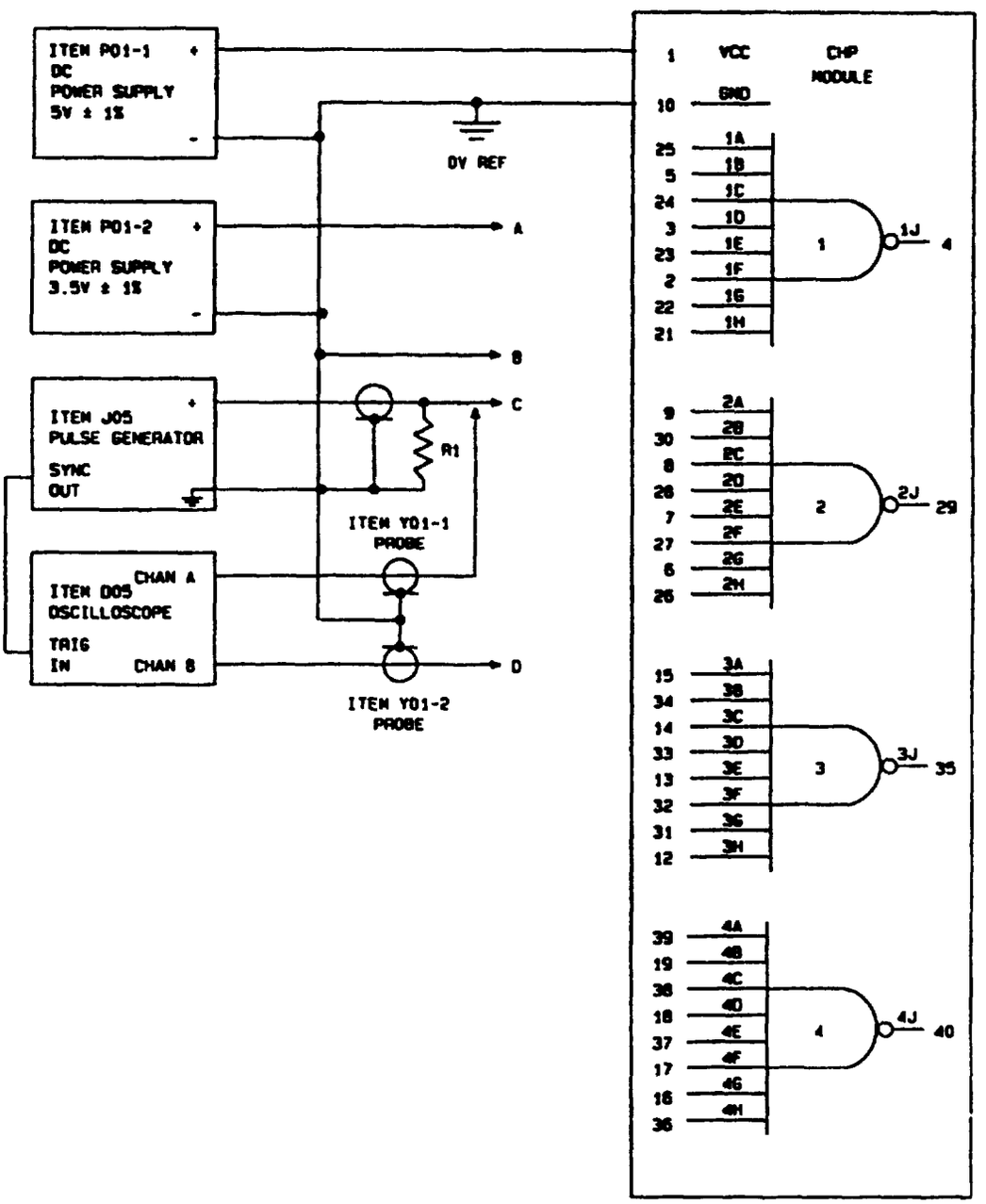
Test cond	Parameter	Connect				Initial limits 25±5°C ambient		Unit	Test fig
		A to pin	B to pin	C to pin	D to pin	Min	Max		
R1	1J Output crosstalk, low level	25,5,24,3, 23,2,22, 21,30,8, 28,7,27,6, 26,34,14, 33,13,32, 31,12,19, 38,18,37, 17,16,36	Open	9, 15, 39	4	0	500	mV	18,19
R2	2J	9,30,8,28, 7,27,6,26, 5,24,3,23, 2,22,21, 34,14,33, 13,32,31, 12,19,38, 18,37,17, 16,36		25, 15, 39	29				
R3	3J	15,34,14, 33,13,32, 31,12,5, 24,3,23,2, 22,21,30, 8,28,7,27, 6,26,19, 38,18,37, 17,16,36		25, 9,39	35				
R4	4J Output crosstalk, low level	39,19,38, 18,37,17, 16,36,5, 24,3,23,2, 22,21,30, 8,28,7,27, 6,26,34, 14,33,13, 32,31,12	Open	25, 9,15	40	0	500	mV	18,19

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TABLE XV. Output crosstalk (low-level and high-level) test - Continued.

Test cond	Parameter	Connect A to pin	Connect B to pin	Connect C to pin	Connect D to pin	Initial limits 25±5°C ambient Min	Initial limits 25±5°C ambient Max	Unit	Test fig
S1	1J Output crosstalk, high level	30,8,28,7, 27,6,26, 34,14,33, 13,32,31, 12,19,38, 18,37,17, 16,36	25, 5, 24, 3, 23, 2, 22, 21	9, 15, 39	4	0	400	mV	18,19
S2	2J	5,24,3,23, 2,22,21, 34,14,33, 13,32,31, 12,19,38, 18,37,17, 16,36	9, 30, 8, 28, 7, 27, 6,26	25, 15, 39	29				
S3	3J	5,24,3,23, 2,22,21, 30,8,28,7, 27,6,26, 19,38,18, 37,17,16, 36	15, 34, 14, 33, 13, 32, 31, 12	25, 9,39	35				
S4	4J Output crosstalk, high level	5,24,3,23, 2,22,21, 30,8,28,7, 27,6,26, 34,14,33, 13,32,31, 12	39, 19, 38, 18, 37, 17, 16, 36	25, 9,15	40	0	400	mV	18,19

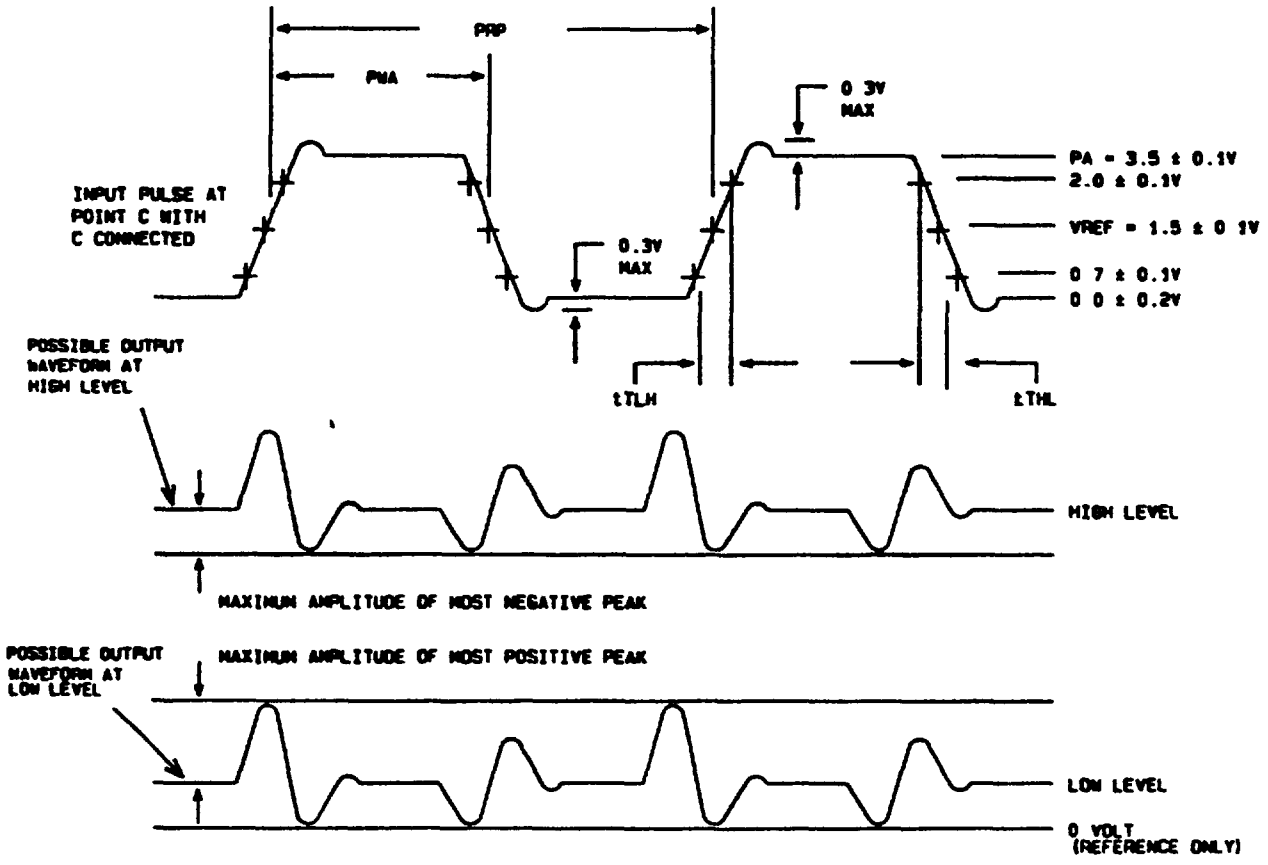
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NOTES:
1. R1 = 51 Ohm ± 5%, 1/2 W.
2. Load all outputs with load as shown on figure 12.

FIGURE 18. Output crosstalk (low-level and high-level) test circuit for table XV.

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NOTES:
1. Input pulse parameters:
   tTLH = tTHL = 3.5 ± 1.5 ns (verified at each input pin).
   PWA = 500 ± 50 ns.
   PRP = 1,000 ± 100 ns.
2. The lead length of electrical conductors between resistor R1 and input pins; between items Y01-1 and Y01-2 and input and output pins; between load circuits and output pins; and between ground leads of items Y01-1 and Y01-2 and test fixture and module ground pins shall be less than 1 inch (25.4 mm).

FIGURE 19. Output crosstalk (low-level and high-level) test circuit waveforms for table XV.

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TABLE XVI. Power supply transient amplitude test.

Test cond	Parameter	Initial limits 25°C±5°C ambient		Unit	Test fig
		Min	Max		
T	Power supply transient amplitude	0	400	mV	20,21

TABLE XVII. Power supply filter capacitance test.

Test cond	Parameter	Connect		Initial limits 25±5°C ambient		Unit	Test fig
		A to pin	B to pin	Min	Max		
U	Power supply filter capacitance	Open	1	1.6	3.0	μF	22

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TABLE XVIII. Input capacitance test.

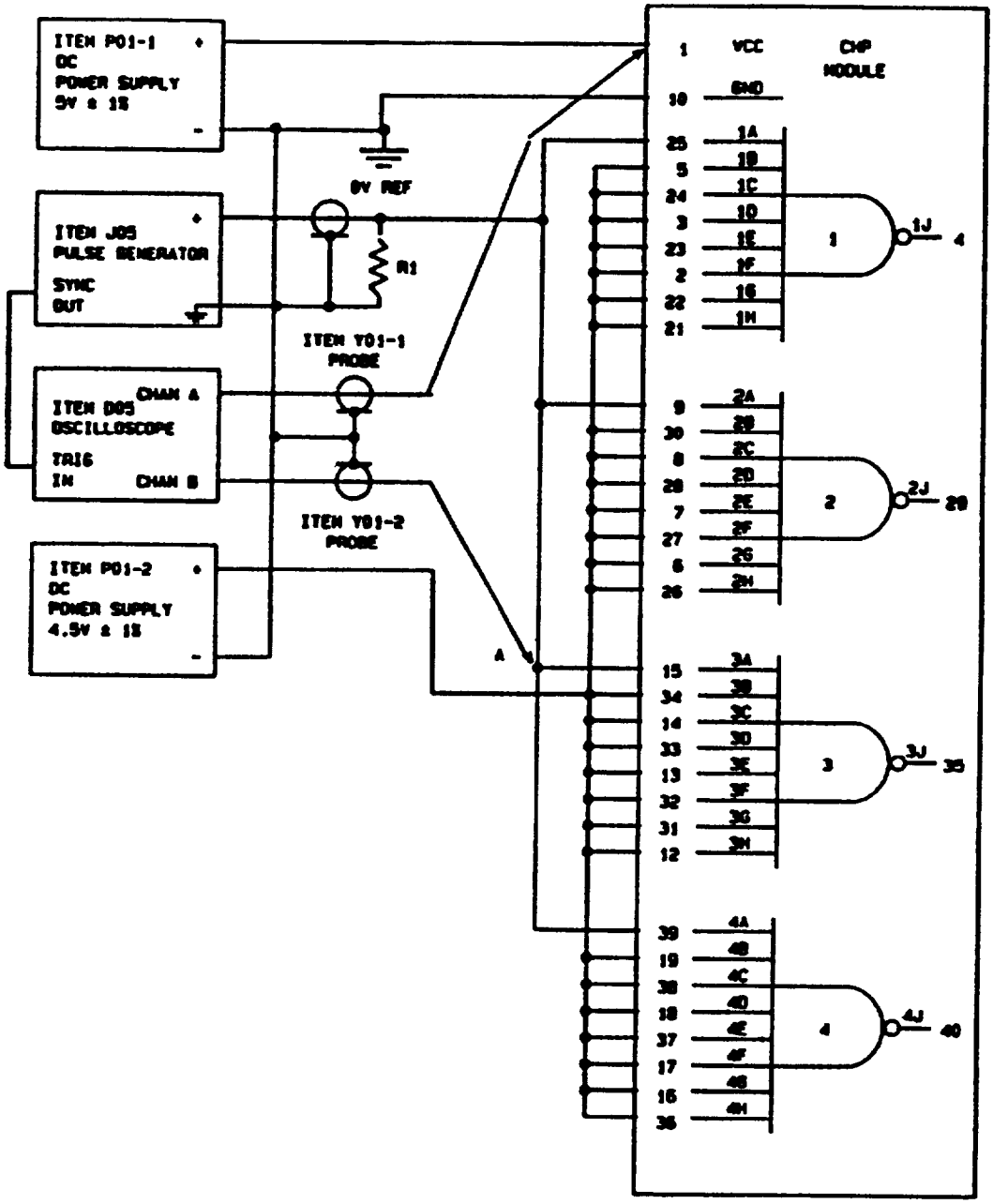
Test cond	Parameter	Connect		Initial limits 25±5°C ambient		Unit	Test fig
		A to pin	B to pin	Min	Max		
V1	1A Input capacitance	25	Open	5	15	pF	22
V2	1B	5					
V3	1C	24					
V4	1D	3					
V5	1E	23					
V6	1F	2					
V7	1G	22					
V8	1H	21					
V9	2A	9					
V10	2B	30					
V11	2C	8					
V12	2D	28					
V13	2E	7					
V14	2F	27					
V15	2G	6					
V16	2H	26					
V17	3A	15					
V18	3B	34					
V19	3C	14					
V20	3D Input capacitance	33	Open	5	15	pF	22

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TABLE XVIII. Input capacitance test - Continued.

Test cond	Parameter	Connect		Initial limits 25±5°C ambient		Unit	Test fig
		A to pin	B to pin	Min	Max		
V21	3E Input capacitance	13	Open	5	15	pF	22
V22	3F	32					
V23	3G	31					
V24	3H	12					
V25	4A	39					
V26	4B	19					
V27	4C	38					
V28	4D	18					
V29	4E	37					
V30	4F	17					
V31	4G	16					
V32	4H Input capacitance	36	Open	5	15	pF	22

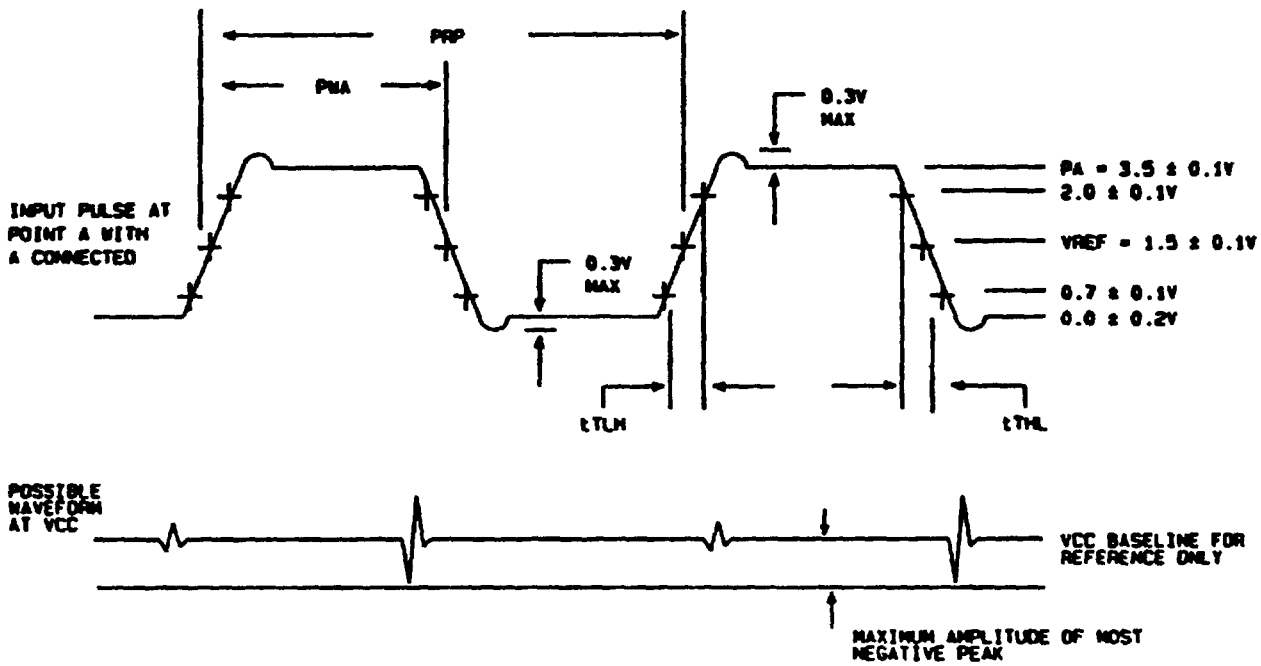
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NOTE: R1 = 51 Ohm ± 5%, 1/2 W.

FIGURE 20. Power supply transient amplitude test circuit for table XVI.

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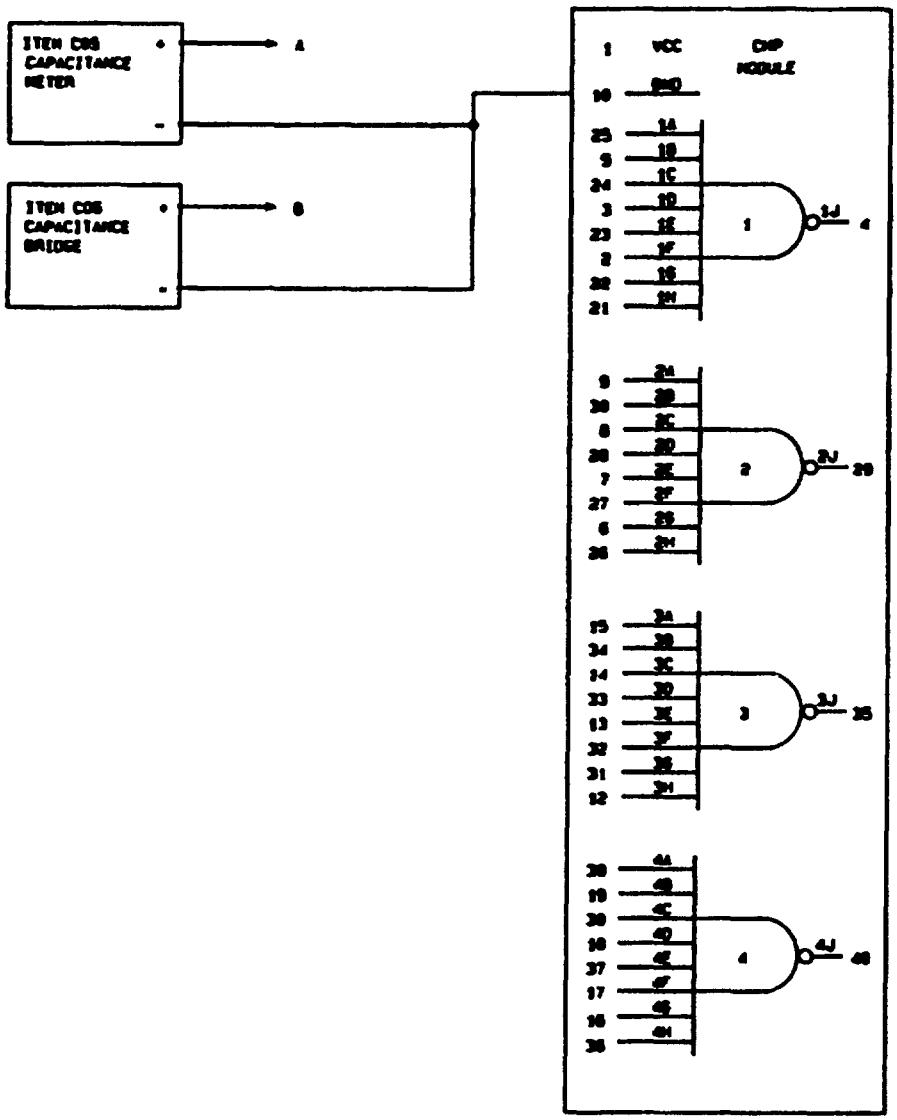


NOTE: Input pulse parameters:
tTLH = tTHL = vary tTLH from 5 to 25 ns and tTHL from 3 to 10 ns and record maximum value of transient amplitude.

PWA = 500 ± 50 ns.
PRP = 1,000 ± 100 ns.

FIGURE 21. Power supply transient amplitude test circuit waveforms for table XVI.

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NOTES:

1. Null out the effects of the test leads prior to connecting the module.

2. Do not attach test connector or wires to any of the module pins other than pin 10 and the pin under test. Stray capacitance added to other pins may affect the readings.

3. Item C06 setup:
   Frequency = 1 KHz
   Vpk = 0.3 V maximum.

*FIGURE 22. Power supply filter capacitance and input capacitance test circuit for tables XVII and XVIII.

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